# Asset Transfers and Self-Fulfilling Runs

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**Abstract**

We introduce a new mechanism that eliminates self-fulfilling runs on a Diamond Dybvig intermediary without requiring deposit insurance. During a run, a depositor can take unliquidated intermediary assets in exchange for closing their account. We interpret our mechanism as a repo contract, covered bond, or bankruptcy plan. We show our mechanism improves upon suspension of convertibility if policy makers have limited commitment, depositors repeatedly interact with the intermediary, and/or there is aggregate return risk.

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# 1 Introduction

The 2007-09 financial crisis and 2023 Silicon Valley Bank collapse showed that bank runs are still a problem. The policy response has been to expand deposit insurance to additional financial institutions and large depositors. However, historical limits on deposit insurance reflect concerns that it encourages inefficient behavior by financial institutions and discourages scrutiny by depositors. We provide an alternative solution to the run problem: a financial intermediary allows depositors to take assets from its balance sheet in exchange for closing their account. In a run, patient depositors take assets, saving the intermediary from costly liquidation. As a result, self-fulfilling runs can no longer be sustained.

We return to the canonical environment for discussing bank runs provided by Diamond and Dybvig (1983). There are two main periods and an intermediary with holdings of an asset that can be liquidated in the earlier period or held until the later period for a positive return. Depositors privately know whether they are impatient and can only consume in the earlier period, or whether they are patient and can consume in either period. Depositors arrive sequentially in the earlier period to interact with the intermediary. In the original demand deposit mechanism, the intermediary offers two options: withdraw goods now or come back in the later period to receive a higher quantity of goods. The intermediary wants to provide liquidity insurance, which effectively allocates more assets to each impatient depositor than to each patient depositor. This creates the possibility of a self-fulfilling run in which patient depositors withdraw goods early because they are concerned others are doing the same and that the intermediary will therefore become insolvent.

We propose an alternative mechanism that allows the intermediary to provide liquidity insurance without introducing the possibility of a run. The intermediary offers depositors the original two options as well as a third, which we call the "asset transfer": immediately taking ownership of an asset from the intermediary's balance sheet.[1] Conceptually, our mechanism says to depositors: if you are worried about a run and don't need to consume immediately, then you can swap your

[1]To our knowledge, asset transfers have not been discussed in the Diamond and Dybvig (1983) literature, but we believe there is nothing in the original environment preventing their use. For a thorough discussion and interpretation of the essential features that characterize the Diamond and Dybvig (1983) environment, see Wallace (1988).

claim on the intermediary for a claim directly on the intermediary's assets. Patient depositors hold assets until the later period and receive the positive return, so this is the efficient way to provide them value while allowing them to end their relationship with the intermediary. Self-fulfilling runs no longer occur because each patient depositor who takes the asset transfer improves the intermediary's balance sheet, which discourages other patient depositors from withdrawing early.

Our model produces real world takeaways for policy makers. First, it is unnecessary and unwise to extend deposit insurance to large sophisticated depositors because they are most able to impose scrutiny on intermediaries and to accept asset transfers during a run.[2] Second, contracts should be set up in advance that allow depositors and financial intermediaries to negotiate over who holds intermediary assets when a run occurs. Third, stress tests and other regulation should consider the feasibility of transferring assets to depositors.

To explore the generality of the asset transfer mechanism, we study a collection of extensions to the Diamond and Dybvig (1983) environment. In Section 3.1, we address the possibility that if intermediaries can transfer assets to depositors, then assets are tradable more generally. This opens up the possibility of "side-trades" where patient depositors withdraw goods early in order to purchase assets from outside agents unaffiliated with the intermediary. Asset transfers are less effective than in our baseline environment, but remain useful for eliminating runs as long as there are some transaction costs and as long as there is sufficient heterogeneity in depositors' returns (net of frictions) on intermediary assets vs. outside assets. This extension suggests a role for intermediary coordination or policy in order to restrict asset sales from one intermediary to depositors at another, or to restrict depositor movement between intermediaries during a crisis.

In Section 3.2, we demonstrate that the asset transfer mechanism is more effective than a suspension mechanism. Suspensions are a much discussed solution to the run problem: if too many depositors withdraw goods in the early period, the intermediary temporarily closes, ensuring there will be goods available for depositors who show up in the later period. We develop three extensions to the baseline environment that highlight potential problems with suspension that do not apply to

[2]Following the recent Silicon Valley Bank run, there were calls to extend deposit insurance to large depositors.

asset transfers: limited commitment, repeat visits to the intermediary during the early period, and aggregate return risk. Intuitively, suspension can be an effective off-equilibrium-path threat, but is costly to depositors if actually implemented. Moreover, it can prevent the efficient liquidation of bad assets. On the other hand, asset transfers work because they are optional for each depositor, encourage the efficient use of assets, and improve the intermediary's balance sheet in a run.

Finally, in Section 3.3, we introduce idiosyncratic return risk within an intermediary as one particular reason why an intermediary may be better equipped than depositors to hold assets, even if its only ability is aggregation. The intermediary can create a fully diversified portfolio, but can only transfer units of the asset that still bear risk. We show that if idiosyncratic return risk is sufficiently large—even without any aggregate risk—then an asset transfer that risk-averse patient depositors prefer over taking goods is too expensive, so asset transfers cannot prevent runs. This is particularly relevant in light of the work by Ospina and Uhlig (2018), which shows that the increase in the average loss rate on mortgage backed securities during the 2007-09 financial crisis was relatively low compared to the increase in the dispersion of loss rates.

*Literature Review:* It is illustrative to compare our paper to Jacklin (1987), which introduces an equity contract that, at first glance, may appear similar to our asset transfer. There are two key differences. First, the equity contract in Jacklin (1987) is a claim on the residual holdings of the intermediary, whereas our asset transfer is effective exactly because it delivers a claim on future goods independent of the intermediary and so ends a depositor's relationship with a failing intermediary. Second, unlike our asset transfer, the equity contract relies on ex-post depositor trade to deliver different allocations to impatient and patient depositors, which constrains those allocations to deliver the same market value (preventing any provision of insurance under the generalized preferences in Jacklin (1987)).

Our paper complements the literature arguing that redesigning financial contracts is a better way to reduce financial instability than expanding deposit insurance and bailouts (for example, Cochrane (2014), Philippon (2016)). Previous work mostly focuses on narrow banking and net

asset value shares in mutual funds, whereas we eliminate runs by setting up contracts that facilitate direct negotiation between depositors and financial intermediaries. One distinction is that our mechanism preserves the role of an intermediary as a provider of liquidity insurance.

Formally, our approach follows a literature, pioneered by Green and Lin (2000, 2003) and Peck and Shell (2003), which studies whether a mechanism designer—interpreted as a competitive financial sector—facing the essential features of the Diamond and Dybvig (1983) "banking" environment can strongly implement the first best allocation. We are most related to the papers in this literature that propose indirect mechanisms to solve the run problem, such as Andolfatto et al. (2017) and Cavalcanti and Monteiro (2016). We make two contributions. First, our mechanism does not require the same level of commitment as do other indirect mechanisms. In Andolfatto et al. (2017), the intermediary must commit to punishing some depositors who choose the third option even though doing so will not be ex-post optimal. In Cavalcanti and Monteiro (2016), the intermediary uses the third option in their indirect mechanism to extract information, from an arbitrarily small collection of depositors, to work out whether to suspend withdrawals. Hence, their mechanism as well as other mechanisms that essentially use suspension, such as in De Nicolo (1996), are subject to the commitment problems described in Ennis and Keister (2009, 2010), which we discuss in Section 3.2 along with other reasons why suspension may not work. Second, our mechanism has a simple real world interpretation that allows us to develop new criteria for whether intermediaries are subject to self-fulfilling runs. Moreover, it suggests a new direction for future research: studying how depositors and intermediaries can set up contracting arrangements to facilitate asset transfers.

# 2 Asset Transfers in Theory

## 2.1 The Environment

We consider the classic Diamond and Dybvig (1983) environment with a sequential service constraint formalized in the manner of Wallace (1988). There are three time periods, $t = 0, 1, 2$, and a

continuum of depositors indexed by $i \in [0,1]$. Each depositor $i$ has preferences given by

$$U(c_{i,1}, c_{i,2}; \theta_i) = \begin{cases} u(c_{i,1}), & \text{if } \theta_i = I \\ u(c_{i,1} + c_{i,2}), & \text{if } \theta_i = P \end{cases}$$

where $c_{i,t}$ is depositor $i$'s consumption in period $t$ of the good, and $\theta_i \in \{I, P\}$ is the depositor's type. If $\theta_i = I$, then depositor $i$ is impatient and only cares about consumption in period 1. If $\theta_i = P$, then depositor $i$ is patient and cares about total consumption across periods 1 and 2. A depositor's type is revealed to them at the beginning of $t = 1$ and is private information. Denote by $\lambda$ the probability that a depositor is impatient. By the law of large numbers, $\lambda$ is also the fraction of depositors who are impatient, so there is no uncertainty about the aggregate type distribution. The function $u$ is twice differentiable, strictly increasing, strictly concave, and has the property that, for all $c \geq 0$, $-cu''(c)/u'(c) > 1$.

Each depositor is endowed with one unit of the good in period 0. Depositors have access to a constant returns to scale investment technology for transforming the endowment into the consumption good in later periods. An investment in period zero yields a return of $R > 1$ units of the good in period 2 per unit of the good invested. If the project is interrupted in period 1 before completion, it yields 1 unit of the good per unit invested. Once interrupted, an investment cannot be restarted. It is useful to think of this investment as generating a perfectly divisible asset that allows the holder, in period 1, to make an irreversible choice between 1 unit of the good per unit of the asset in period 1 and $R$ units of the good per unit of the asset in period 2. If the holder chooses to receive goods in period 1, then we say the asset has been liquidated.

There is also an intermediary in which depositors can pool resources to manage liquidity risk. In period 0, endowments are deposited and invested, which generates an intermediary balance sheet in period 1 consisting of a unit measure of the asset described above. In period 1, depositors cannot interact with each other and instead contact the intermediary. Upon contact, the intermediary offers a menu of options. In period 2, depositors can interact with each other and the intermediary freely.

Interaction in period 1 is as follows. Each depositor contacts the intermediary exactly once.[3] They do so one at a time: at the beginning of period 1, each depositor draws a place in line $s$ from the uniform distribution on $[0,1]$ independent of their type, where $s$ represents the proportion of depositors ahead of them. When a depositor interacts with the intermediary, all they observe are the options offered. In particular, they neither observe their own place in line nor the actions of previous depositors. The depositor selects one of the options and the intermediary can make an immediate transfer. The depositor and intermediary do not interact again until the following period. These restrictions placed on the intermediary are called the "sequential service constraint".

The intermediary can transfer goods and/or assets to depositors in period 1, and goods in period 2.[4] Asset transfers are subject to a per unit cost: if the intermediary transfers 1 unit of the asset, then the depositor receives $1-\varepsilon$ units. An alternative interpretation is that if the intermediary holds units of the asset from period 1 to 2, then it receives return $R$ units of the good, and if a depositor holds units of the asset, then it receives return $R^a \leq R$, where $R^a := (1-\varepsilon)R$. We view previous papers in the literature (that do not consider asset transfers) as implicitly setting $\varepsilon = 1$. Throughout the rest of the paper, for convenience, we refer to the effective depositor return $R^a$ rather than the transfer cost $\varepsilon$.

## 2.2 Unconstrained Social Planner

Suppose there is a benevolent social planner who observes depositor types and directly controls allocations. The social planner maximizes welfare subject to the aggregate resource constraint, where welfare is depositors' period 0 expected utility. Diamond and Dybvig (1983) show that the first best allocation satisfies:

[3] We can also allow each depositor to choose not to contact the intermediary in period 1 when it is their turn.

[4] There are no assets in period 2.

$$(c_{i,1}, c_{i,2}) = \begin{cases} (c_1^*, 0), & \text{if } \theta_i = I \\ (0, c_2^*), & \text{if } \theta_i = P \end{cases}$$

where $c_1^*$ and $c_2^*$ satisfy the Euler equation $u'(c_1^*) = Ru'(c_2^*)$ and the aggregate resource constraint $(1-\lambda)c_2^* = R(1-\lambda c_1^*)$. Since $R > 1$ and the coefficient of relative risk aversion is always strictly greater than 1, it follows that $1 < c_1^* < c_2^* < R$. These inequalities demonstrate that the planner provides insurance for agents' type risk, which can be interpreted as liquidity insurance.

## 2.3 Constrained Social Planner

Following Green and Lin (2000, 2003), we use a mechanism design approach to investigate which outcomes can be achieved by a constrained planner in the environment described in Section 2.1. The constrained planner faces the sequential service constraint and an information asymmetry about each depositor's type.

A depositor's type consists of their patience type $\theta \in \{I, P\}$, which they know but the planner does not, and their place in line $s$, which the planner knows but they do not. An outcome function specifies that the depositor at position $s$ gets allocation

$$(c_1(a_s, s; \mathbf{a}), c_2(a_s, s; \mathbf{a}), \kappa(a_s, s; \mathbf{a})),$$

where $a_s$ is the action of the depositor at place $s$ in line, $\mathbf{a}$ is the vector of all depositors' actions, $c_t(a_s, s; \mathbf{a})$ is the units of the consumption good given to the depositor in period $t$, and $\kappa(a_s, s; \mathbf{a})$ is the units of the asset the intermediary transfers in period 1.[5] A mechanism consists of a set of actions available to each depositor, and an outcome function. The constrained social planner is restricted to choose a sequential service feasible mechanism:

[5] These are the only three objects in the economy, so this is as general an outcome space as possible.

**Definition 1** (**Sequential Service Feasible Mechanism**). Let $B(s;\mathbf{a})$ denote the measure of the intermediary's holdings of the asset at place $s$ if depositors play $\mathbf{a}$. A mechanism is sequential service feasible if for all action profiles $\mathbf{a}$, the outcome function satisfies:

1. Budget feasibility: $\int_0^1 c_2(a_s,s;\mathbf{a})ds \leq RB(1;\mathbf{a})$

2. Sequential service constraint: The period 1 payouts to the depositor at place $s$ in line, $c_1(a_s,s;\mathbf{a})$ and $\kappa(a_s,s;\mathbf{a})$, can only depend on $s$ itself and $\{a_l : l \leq s\}$, the actions of depositors at place up to and including $s$.

The mechanism induces the following game. Each depositor chooses a mixed strategy that can depend on whether they are patient, $\theta \in \{I,P\}$, and the options they are offered, but not explicitly on the actions of previous depositors or on their place in line, $s$. The equilibrium concept is Bayes Nash: depositors maximize their expected utility, taking $\mathbf{a}$ as given since they have measure zero. We focus on the implementation of social choice functions in which each depositor's allocation depends only on whether they are impatient or patient, i.e., their allocation is given by $h : \{I,P\} \to \mathbb{R}^3_+$. We say a mechanism **weakly implements** the social choice function $h$ if there exists a Bayes Nash equilibrium of the game induced by the mechanism in which impatient depositors receive $h(I)$ and patient depositors receive $h(P)$. A mechanism **strongly implements** if in every Bayes Nash equilibrium, depositor allocations are given by $h$.

The canonical question in the Diamond and Dybvig (1983) literature can be phrased as: does there exist a sequential service feasible mechanism that strongly implements the first best allocation? If this is not the case, then the environment is fundamentally unstable since we cannot achieve the first best without introducing the possibility of suboptimal equilibria.

## 2.4 Demand Deposit Mechanism

Diamond and Dybvig (1983) and most of the subsequent literature study the demand deposit mechanism, which is a direct mechanism.[6] Each depositor chooses from {withdraw, wait}, where "with-

[6]Notable exceptions are Andolfatto et al. (2017) and Cavalcanti and Monteiro (2016), who also propose indirect mechanisms.

draw" represents withdrawing goods in period 1 and "wait" represents waiting to withdraw goods in period 2.[7] The outcome function is

$$\begin{aligned}
&\left(c_1^D(a_s,s;\mathbf{a}), c_2^D(a_s,s;\mathbf{a}), \kappa^D(a_s,s;\mathbf{a})\right) \\
&\qquad = \begin{cases} \begin{bmatrix} (c_1^*, 0, 0), & \text{if } B(s;\mathbf{a}) > 0 \\ (0,0,0), & \text{if } B(s;\mathbf{a}) = 0 \end{bmatrix}, & \text{if } a_s = \text{withdraw} \\ (0, RB(1;\mathbf{a})/m_0, 0), & \text{if } a_s = \text{wait} \end{cases}
\end{aligned}$$

where $m_0$ is the fraction of depositors who choose "wait" and

$$B(s;\mathbf{a}) = 1 - \int_0^s c_1^D(a_s,s;\mathbf{a})ds$$

is the measure of the intermediary's holdings of the asset remaining at place $s$. In words, each depositor can withdraw $c_1^*$ goods in period 1 as long as the intermediary has any assets remaining. If a depositor instead waits, then they split the return on the intermediary's balance sheet with all other waiting depositors in period 2.

Diamond and Dybvig (1983) prove that the mechanism is sequential service feasible and weakly implements the first best allocation. More specifically, they show that the induced game has a "truth telling" equilibrium as well as a "run" equilibrium. In the former, impatient depositors withdraw in period 1, patient depositors wait, and the allocation is the first best. In the run, all depositors withdraw in period 1, and the intermediary runs out of resources before the end of the line, i.e., there exists an $\bar{s} < 1$ such that $B(\bar{s};\mathbf{a}) = 0$ and so all depositors in places $s \geq \bar{s}$ receive no goods in either period.

[7] Technically, Diamond and Dybvig (1983) allow depositors to withdraw a fraction in period 1. However, this is equivalent to the setup here because we have a continuum of depositors and allow for mixed strategies.

## 2.5 Asset Transfer Mechanism

We define an asset transfer mechanism that makes use of the intermediary's ability to transfer ownership of units of the asset. Each depositor chooses from three actions, $\{\text{withdraw}, \text{wait}, \text{transfer}\}$. The first two actions, "withdraw" and "wait", are as in the demand deposit mechanism. If a depositor chooses the new action, "transfer", then the intermediary transfers $\kappa$ units of the asset (not liquidated), where $\kappa$ is a single number.[8] Thus, the asset transfer mechanism is like the demand deposit mechanism except now if a depositor chooses not to wait, then they can withdraw $c_1^*$ goods *or* take $\kappa$ unliquidated assets, as long as the intermediary has any asset remaining. A distinct feature of the asset transfer is that a depositor must then choose whether to liquidate those assets or hold them to maturity; by contrast, if a depositor withdraws goods or waits, then they must immediately consume goods upon receipt. If an impatient depositor takes the asset transfer, then they optimally liquidate because they do not value goods in period 2. If a patient depositor takes the asset transfer, then they hold the assets to maturity because the return is $R > 1$.

The outcome function for the asset transfer mechanism is

$$\left(c_1^K(a_s, s; \mathbf{a}), c_2^K(a_s, s; \mathbf{a}), \kappa^K(a_s, s; \mathbf{a})\right)$$

$$= \begin{cases} \begin{bmatrix} (c_1^*, 0, 0), & \text{if } B(s;\mathbf{a}) > 0 \\ (0, 0, 0), & \text{if } B(s;\mathbf{a}) = 0 \end{bmatrix}, & \text{if } a_s = \text{withdraw} \\ \begin{bmatrix} (0, 0, \kappa), & \text{if } B(s;\mathbf{a}) > 0 \\ (0, 0, 0), & \text{if } B(s;\mathbf{a}) = 0 \end{bmatrix}, & \text{if } a_s = \text{transfer} \\ (0, RB(1;\mathbf{a})/m_0, 0), & \text{if } a_s = \text{wait} \end{cases}$$

[8] In Appendix A, we prove Lemma 1, which shows that restricting $\kappa$ to be a single number does not affect whether the first best is strongly implementable.

where $m_0$ is again the fraction of depositors who choose "wait" and

$$B(s;\mathbf{a}) = 1 - \int_0^s c_1^K(a_s, s;\mathbf{a})ds - \int_0^s \kappa^K(a_s, s;\mathbf{a})ds$$

is the measure of the intermediary's holdings of the asset left at place $s$. By construction, the mechanism is sequential service feasible. The following theorem, our main result, characterizes when we can design the mechanism so that it is not subject to the same run problem as the demand deposit mechanism.

**Theorem 1.** *The asset transfer mechanism can strongly implement the first best allocation if and only if* $R^a > \frac{R}{c_2^*/c_1^*}$.

*Proof.* See Appendix A. □

To understand the theorem, it is helpful to recall why a run can occur in the demand deposit mechanism. If impatient depositors withdraw goods in period 1 and patient depositors wait until period 2, then the intermediary's asset holdings are exactly enough to deliver the first best allocation. In this case, which we call "truth telling", the intermediary provides insurance by liquidating $c_1^* > 1$ units of the asset for each impatient depositor and keeping the remaining assets, $1 - \lambda c_1^*$, to provide $c_2^* < R$ units of the good to each patient depositor in period 2. On the other hand, if a sufficiently large fraction of patient depositors withdraw goods in period 1, then the intermediary's asset holdings are depleted before all depositors are served in period 1 because $c_1^* > 1$. In that case, any depositors who wait until period 2 receive nothing. Thus, it is not incentive compatible for a patient depositor to wait if sufficiently many other patient depositors withdraw early.

Now consider the asset transfer mechanism when the inequality in Theorem 1 holds, i.e., $R^a$ is sufficiently high so that the intermediary can choose an asset transfer $\kappa$ that satisfies $R^a\kappa > c_1^*$ and $\kappa \leq c_2^*/R$. For any depositor strategy profile, all impatient depositors strictly prefer to withdraw goods in period 1, and all patient depositors strictly prefer to take the asset transfer over withdrawing goods in period 1. Hence, the only possible incentive compatible deviation from truth telling is for patient depositors to take the asset transfer. Since $\kappa \leq c_2^*/R$, if a positive measure of

patient depositors do so, then unlike if they were to withdraw goods, the intermediary is left at the end of period 1 with weakly *more* units of the asset per patient depositor who chose to wait. In this sense, the asset transfer is an off-equilibrium path option that, by strictly dominating withdrawing goods early for patient depositors, ensures it is always incentive compatible for patient depositors to wait until period 2. As such, the asset transfer mechanism, which is indirect, is able to strongly implement the first best while the demand deposit mechanism, which is direct, cannot.[9]

If $R^a = R$ or equivalently, $\varepsilon = 0$, then a *direct* mechanism can strongly implement the first best: eliminate the option to wait and set $\kappa = c_2^*/R^a$ so that patient depositors take the asset transfer in equilibrium. Intuitively, if the asset transfer is costless, then in period 1, the intermediary can conclude its relationships with *all* depositors by giving units of the good to impatient depositors and units of the asset to patient depositors.[10]

To build further intuition for the result, it is useful to think about two features that a third option (like the asset transfer) must satisfy to eliminate the run. First, for patient depositors, the third option must strictly dominate withdrawing goods. This concerns the priority of different claims. In the demand deposit mechanism, those who withdraw get a claim that is handed out immediately with fixed value $c_1^*$, so their priority is higher than that of all depositors who have not yet been served. Those who wait get a claim that is essentially equity—they receive a share of whatever is left in period 2—so their priority is lower than that of *everyone who withdraws*. The asset transfer is effective because it delivers the same priority as withdrawing goods: a fixed quantity of assets, $\kappa$, are handed out immediately. By contrast, we can imagine an alternative third option that delivers a claim with a fixed payment in period 2, $c_2^*$, as long as the intermediary has goods available, i.e., with lower priority than period 1 goods withdrawals but higher priority than equity. This third option *would not prevent the run* because if everyone withdraws in period 1, then it still delivers

[9] In this context, the revelation principle does not imply that restricting to direct mechanisms is without loss of generality. As discussed in Palfrey (1993), it is possible that an indirect mechanism strongly implements the first best whereas a direct mechanism only weakly implements the first best.

[10] The intermediary has two options, transferring the asset immediately and transferring the good immediately, one of which is relatively more valuable to patient depositors, the other of which is relatively more valuable to impatient depositors, and neither of which has a value dependent on the actions of other depositors. With these two options, the intermediary can separate impatient and patient depositors into their first best allocations without creating the possibility of a run.

nothing. In other words, in a run, getting higher priority than equity is worthless. For a third option to strictly dominate withdrawing goods from the perspective of patient depositors, it must deliver higher priority than the claims of all future depositors—of those not yet served in period 1 and of those who will return to the intermediary in period 2.

The second necessary feature for the third option to eliminate the run is that it uses fewer period 1 intermediary assets than the first best patient depositor allocation. Specifically, a patient depositor's first best allocation uses $c_2^*/R$ assets and the goods withdrawal uses $c_1^* > c_2^*/R$ assets. This creates the possibility of a self-fulfilling run in the demand deposit mechanism because it means patient depositors' choices to withdraw early (run) are strategic *complements*: the more patient depositors withdraw, the *less* remains for each depositor who waits, so the more patient depositors want to withdraw rather than wait. On the other hand, the asset transfer is effective if it uses less than $c_2^*/R$ assets. In that case, patient depositors' choices to take the asset transfer are strategic *substitutes*: the more patient depositors take the asset transfer, the *more* remains for each depositor who waits, so the more patient depositors want to wait rather than take the asset transfer.

Now we can interpret Theorem 1 through the lens of these two features. First, the asset transfer delivers sufficiently high priority, but this priority is only valuable if it also delivers more goods to a patient depositor than withdrawing: $R^a\kappa > c_1^*$. Second, the asset transfer must use less assets than a patient depositor's first best allocation: $\kappa \leq c_2^*/R$. The intermediary can choose a $\kappa$ that satisfies both if and only if the inequality in the theorem holds.

Finally, we show that even if the asset transfer mechanism cannot strongly implement the first best, the condition on the transferred asset return, $R^a$, is weaker for asset transfers to be useful. Specifically, consider allocations $(\bar{c}_1, \bar{c}_2)$ that are feasible and provide insurance: impatient depositors consume $\bar{c}_1$, patient depositors consume $\bar{c}_2$, the allocation satisfies the resource constraint, and $1 < \bar{c}_1 < \bar{c}_2 < R$. The first best is one of these allocations, but there is a continuum. The demand deposit mechanism cannot strongly implement *any* such allocation. The following corollary shows when the asset transfer mechanism can strongly implement such an allocation.

**Corollary 1.** *Suppose an allocation $(\bar{c}_1, \bar{c}_2)$ is feasible and provides insurance. The demand deposit mechanism can weakly implement $(\bar{c}_1, \bar{c}_2)$, and there is an equilibrium in which all depositors run. The asset transfer mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if $R^a > \frac{R}{\bar{c}_2/\bar{c}_1}$.*

The corollary states that asset transfers are useful if and only if the return on transferred assets is sufficiently high ($R^a > 1$) or equivalently, the transfer cost is sufficiently low ($\varepsilon < 1 - 1/R$). In this case, the asset transfer mechanism can strongly implement some insurance, i.e., an allocation $(\bar{c}_1, \bar{c}_2)$ with $\bar{c}_2/\bar{c}_1 < R$. Intuitively, $R^a > 1$ means it is more efficient to provide patient depositors with value by transferring assets than by liquidating assets. Furthermore, the more insurance the intermediary wants to strongly implement using asset transfers—the lower is $\bar{c}_2/\bar{c}_1$—the higher the required $R^a$. The first best entails the maximum desirable insurance.

## 2.6 Real World Interpretations

As we just discussed, the key feature of asset transfers is that they offer a high priority claim to patient depositors without liquidating assets. We now discuss possible real world interpretations of the asset transfer, i.e., real world ways in which an intermediary can create such claims.

*Repurchase agreement ("repo" contract):* There are two ways to interpret our asset transfer mechanism as a repo contract. First, we can think of the entire mechanism as a repo contract. Specifically, in period 0, each depositor keeps $\kappa$ units of the asset and gives $1 - \kappa$ units to the intermediary. In our asset transfer mechanism, a depositor can choose withdraw, wait, or asset transfer in period 1. To map this into a repo contract, we can think of each option as follows. If a depositor withdraws, then the intermediary repurchases the depositor's $\kappa$ holdings of the asset using $c_1^*$ period 1 goods. If a depositor chooses to wait, then the intermediary repurchases the depositor's $\kappa$ asset holdings using a claim to $c_2^*$ period 2 goods.[11] Finally, if a depositor takes the asset transfer, then

[11] This claim can look more like debt or like equity. That is, for the proofs of Theorem 1 and Corollary 1, it does not matter what the intermediary does with excess period 2 goods beyond what's necessary to give $c_2^*$ (or $\bar{c}_2$) to each waiting depositor. Moreover, it does not matter how the intermediary disperses period 2 goods when it does not have enough to give $c_2^*$ (or $\bar{c}_2$) to each waiting depositor.

they end the repurchase agreement and walk away with their $\kappa$ units of the asset.

Alternatively, we can think of the repo contract as only used if a depositor takes the asset transfer. Specifically, as in our original setup, each depositor gives their entire endowment to the intermediary in period 0. In period 1, a depositor can choose withdraw, wait, or repo, where the repo option takes the place of the asset transfer option in our original mechanism. If a depositor chooses repo, then the intermediary gives them $\kappa$ units of the asset with the right, at the end of period 1, to sell it back to the intermediary in exchange for the same period 2 claim held by those who chose to wait. So if a patient depositor chooses repo and it turns out there is a run, then they will keep the $\kappa$ units of the asset, hold them to maturity, and consume the proceeds in period 2. If it turns out there is not a run, then the depositor will trigger the option to sell the assets back to the intermediary, and will receive goods in period 2 as if they had originally waited.

*Covered bond with earmarked collateral:* We can also interpret the asset transfer as offering a covered bond. Specifically, replace the asset transfer option in our original mechanism with a covered bond option. For each depositor that chooses the covered bond in period 1, $\kappa$ units of the asset are added to a pool of "earmarked" or "ring-fenced" high priority collateral. In period 2, the proceeds on a depositor's earmarked collateral are $R^a\kappa$ goods. If the intermediary has enough period 2 goods so that all covered bond holders and waiting depositors can get at least $R^a\kappa$, then the earmarked pool reenters the intermediary's general balance sheet, and all covered bond holders and waiting depositors get an even share of the total proceeds. On the other hand, if the intermediary does not have enough period 2 goods, then covered bond holders each get $R^a\kappa$ goods funded by the earmarked collateral, and any remaining goods are split evenly among waiting depositors. Importantly, earmarked assets cannot be used to fund goods withdrawals in period 1, so the covered bond offers sufficiently high priority to prevent runs.

*Bankruptcy plan:* Finally, we can interpret the asset transfer as part of a bankruptcy plan that gives depositors the option to progressively disassemble the intermediary if they believe it will become

insolvent. This can be seen as an extension to the arguments in Calomiris and Kahn (1991) in the sense that giving depositors the option to force a dismantling of the bank is more valuable than giving depositors the option to run on the bank.

Some of these interpretations suggest that implementing the asset transfer requires organization at period 0 to securitize the intermediary's assets or to create a legal structure for a bankruptcy plan. Thus an intermediary that offers the asset transfer mechanism and prevents runs may use some of its initial asset holdings to do so, which means even in the good equilibrium (where impatient depositors withdraw early and patient depositors wait), it provides impatient depositors less than $c_1^*$ and patient depositors less than $c_2^*$. In that case, we may envision that depositors choose between run prone intermediaries that offer high payouts $c_1^*$ and $c_2^*$ in the good equilibrium and intermediaries that are not run prone but offer lower payouts in the good equilibrium.

Finally, depending on the interpretation, it may make sense to think of the asset transfer cost as a transaction cost, a bankruptcy cost, a penalty cost for defaulting on a repo contract, or as capturing that depositors are less efficient holders of the asset.

# 3 Model Extensions / Robustness

In this final section, we conduct exercises to illustrate further when asset transfers are effective, and how they compare to other solutions to the self-fulfilling run problem. We first show the extent to which asset transfers remain effective in the presence of asset trade. Second, we formalize extensions to our environment in which asset transfers outperform suspension mechanisms. Third, we show that idiosyncratic return risk on investments can limit the usefulness of asset transfers.

## 3.1 Asset Trade

We now discuss a limitation on when asset transfers are useful. If the intermediary can transfer assets to its depositors, then assets may be tradable more generally. Going back to Jacklin (1987),

it is known that asset trade may create a problem distinct from runs: it unwinds *any* equilibrium in which the intermediary provides liquidity insurance. Intuitively, providing liquidity insurance means the intermediary uses more assets for each impatient depositor than for each patient depositor. Absent asset trade, this allocation is incentive compatible for patient depositors because the assets used for impatient depositors are irreversibly liquidated in period 1, whereas the assets used for patient depositors are held to maturity until period 2 and generate return $R > 1$. Thus, if patient depositors can essentially undo liquidation by using period 1 goods to purchase assets, then they prefer to deviate and take the impatient depositor allocation, *even if other patient depositors wait*. This problem is therefore different from the classic run problem in that it is about weak rather than strong implementation, i.e., whether the intermediary can provide liquidity insurance in *any* equilibrium rather than as the *unique* equilibrium outcome. Moreover, if an intermediary cannot even weakly implement liquidity insurance, then it should fully unwind in the sense that there is no reason to initially deposit resources in period 0.

To study asset trade in a world with asset transfers, we extend our model by adding outside financial institutions with assets to sell. Depositors can use early goods withdrawals to purchase these assets, potentially subject to frictions similar to the frictions on asset transfers. We ask two questions. First, do asset transfers help with the weak implementation Jacklin (1987) problem caused by asset trade? The answer is no. Regardless of asset trade and transfer frictions, introducing asset trade imposes the same limit on the intermediary's ability to weakly implement liquidity insurance under both the demand deposit and asset transfer mechanisms (Propositions 1 and 2).

Second, we ask whether introducing asset trade reduces the effectiveness of asset *transfers* as a solution to the run (strong implementation) problem. The answer is it depends. Asset transfers can no longer prevent any runs if all depositors get the exact same return (net of any frictions) on assets purchased from outside institutions as on assets transferred from the intermediary (Proposition 1). This is because asset trade increases the value of withdrawing goods early, which raises the cost of providing a useful asset transfer. On the other hand, asset transfers can prevent runs as long as there is sufficient heterogeneity across depositors in their return on (or preference for) intermediary

assets vs. outside assets (Proposition 2). Specifically, as long as *some* depositors get a higher return on transferred intermediary assets than on purchased outside assets, even if, on average, depositors prefer outside assets.

*Environment changes:* Formally, we extend the model as follows. At each place in line in period 1, there is a large number of outside agents (financial institutions) with assets. These outside agents need goods in period 1, so they liquidate their assets or sell them if they can get a better price. Their assets, like the intermediary's, are perfectly divisible and can be irreversibly liquidated for one good per asset in period 1 or held to maturity for $R$ goods per asset in period 2. Each depositor can trade with these outside agents at their place in line. We start by imposing "homogeneous frictions" in the sense that all asset trades are subject to the same per unit cost $\varepsilon$ as asset transfers, so if an outside agent sells $x$ assets to a depositor, then the depositor receives $(1-\varepsilon)x$ assets. As in our baseline model, we instead refer to the effective return $R^a \equiv (1-\varepsilon)R$ that a patient depositor earns on purchased outside assets or transferred intermediary assets. Finally, we suppose trade is efficient in the sense that in equilibrium at each place in line, a trade occurs if and only if it makes both the depositor and outside agents better off.[12]

The following proposition characterizes how much insurance the intermediary can weakly or strongly implement in this homogeneous frictions case. As in Corollary 1, we focus on allocations of the same form as the first best: impatient depositors consume $\bar{c}_1$, patient depositors consume $\bar{c}_2$, the allocation is feasible (satisfies the resource constraint), and the allocation provides insurance, which means $1 < \bar{c}_1 < \bar{c}_2 < R$. Thus, $\bar{c}_2/\bar{c}_1 \in (1,R)$ and more insurance means a lower $\bar{c}_2/\bar{c}_1$.

**Proposition 1.** *Suppose allocation $(\bar{c}_1,\bar{c}_2)$ is feasible and provides insurance. In the homogeneous frictions case, the demand deposit and asset transfer mechanisms can weakly implement $(\bar{c}_1,\bar{c}_2)$ if and only if $\bar{c}_2/\bar{c}_1 \geq R^a$ and cannot strongly implement $(\bar{c}_1,\bar{c}_2)$.*

[12] A depositor's choice of which option to take and of whether to trade occur simultaneously, so their outside option is whichever they prefer out of withdrawing goods for consumption, taking the asset transfer, and waiting. Outside agents' alternative is to liquidate assets. How the surplus is split in a trade does not affect the results that follow, so trade is efficient under Nash bargaining with any bargaining weights. If the surplus is exactly 0, then a trade can occur but does not have to.

*Proof.* See Appendix B. □

The proposition first states that asset trade restricts how much insurance the intermediary can weakly implement: if $\bar{c}_1$ and $\bar{c}_2$ are too close together, then the "good" equilibrium—where impatient depositors withdraw $\bar{c}_1$ goods and patient depositors wait to get $\bar{c}_2$ goods—does not exist. By contrast, in our baseline model without asset trade, the "good" equilibrium exists for any $\bar{c}_2/\bar{c}_1 \in (1,R)$ in both the demand deposit and asset transfer mechanisms. This new weak implementation constraint is the Jacklin (1987) problem mapped into our setting. Our result shows that asset transfers do not help with this problem: the demand deposit and asset transfer mechanisms face the same constraint on what they can weakly implement. Finally, the proposition states that in this homogeneous frictions case, asset transfers no longer help with the classic run (strong implementation) problem: the intermediary cannot strongly implement any liquidity insurance.

Intuitively, asset trade constrains the intermediary by raising the value of withdrawing goods as a patient depositor. That is, if a patient depositor uses $\bar{c}_1$ period 1 goods to purchase assets, then they can get as much as $R^a\bar{c}_1$ period 2 goods because outside agents are willing to sell assets for a price as low as one good per asset. One consequence is a constraint on what the intermediary can weakly implement. That is, if $\bar{c}_2/\bar{c}_1 < R^a$, then patient depositors prefer not to wait *in any equilibrium* because they get more value from withdrawing goods in period 1 to purchase assets than from waiting to withdraw in period 2. A second consequence is that asset transfers no longer prevent runs. Specifically, the early goods withdrawal now generates the same return $R^a$—through trade—as an asset transfer. As such, the intermediary cannot cheaply provide an asset transfer that patient depositors prefer over withdrawing goods early.

*Extension with heterogeneous frictions:* The environment thus far is stark in the sense that asset transfers and trades face the exact same frictions. So we now consider a "heterogeneous frictions" case in which a fraction $\alpha$ of depositors face a lower per unit cost, $\varepsilon_H \in [0,\varepsilon]$, on asset transfers than on asset trades so they face a higher effective return, $R^a_H \equiv (1-\varepsilon_H)R \geq R^a$, on transferred intermediary assets than on purchased outside assets (subscript $H$ for high return). The remaining

fraction $1-\alpha$ of depositors face a higher per unit cost, $\varepsilon_L \geq \varepsilon$, on asset transfers than on asset trades so they face a lower effective return, $R_L^a \equiv (1-\varepsilon_L)R \leq R^a$, on transferred intermediary assets than on purchased outside assets (subscript $L$ for low return). A depositor's return on transferred intermediary assets is realized at the beginning of period 1, is their private and unverifiable information, and is independent of their place in line and of whether they are patient. We add heterogeneity to depositors' returns on intermediary assets rather than on outside assets because it is simpler.[13]

The following proposition characterizes how much liquidity insurance the intermediary can weakly or strongly implement in this heterogeneous frictions case. We continue to focus on allocations of the same form as the first best, so a depositor's allocation does not depend on whether they face a high or low return on transferred intermediary assets.

**Proposition 2.** *Suppose allocation* $(\bar{c}_1, \bar{c}_2)$ *is feasible and provides insurance. In the heterogeneous frictions case, the asset transfer mechanism can weakly implement* $(\bar{c}_1, \bar{c}_2)$ *if and only if* $\bar{c}_2/\bar{c}_1 \geq R^a$ *and can strongly implement* $(\bar{c}_1, \bar{c}_2)$ *if and only if* $\bar{c}_2/\bar{c}_1 > (\alpha R^a/R_H^a + 1 - \alpha)R$*, which is strictly less than* $R$ *if* $\alpha > 0$ *and* $R_H^a > R^a$*.*

*Proof.* See Appendix B. □

The proposition first states that asset transfers do not help with the Jacklin (1987) weak implementation problem, as in the homogeneous frictions case. Specifically, the asset transfer mechanism can weakly implement the same amount of insurance as the demand deposit mechanism.[14] More significantly, the proposition next states that asset transfers are now useful for solving the run problem, i.e., the asset transfer mechanism can strongly implement some liquidity insurance. Intuitively, with heterogeneous frictions, a share $\alpha$ of patient depositors get a higher return from transferred intermediary assets than from period 1 goods used to purchase assets. Any share $\alpha > 0$ and any return advantage over outside assets $R_H^a > R^a$ is sufficient for asset transfers to be useful—some insurance is strongly implementable—because they can create space on the intermediary's

[13] With heterogeneity on outside asset returns, the result is complicated by two types of run equilibrium: one where all patient depositors run and one where they run conditional on getting a high return on outside assets.

[14] The demand deposit mechanism is the same as in the homogeneous frictions case covered by Proposition 1.

balance sheet to withstand a run by the share $1-\alpha$ of patient depositors who get a higher return on outside assets than on intermediary assets.

*Discussion:* Taking Propositions 1 and 2 together, asset transfers do not alleviate the Jacklin (1987) weak implementation problem that asset trade unwinds liquidity insurance. Thus, asset trade cannot be frictionless or else intermediaries cannot improve on autarky. On the other hand, asset transfers remain a useful, though weakened, tool for fighting the run problem as long as depositors face heterogeneous returns on transferred intermediary assets relative to on purchased outside assets. This heterogeneity may reflect differences in asset types across financial institutions along with differences in preferences for these asset types across depositors.

Finally, our results show that asset transfers can be useful for preventing runs as long as some depositors prefer transferred intermediary assets over purchased outside assets. This can emerge from heterogeneous depositor returns/preferences, as in our heterogeneous frictions extension, but it can also occur if depositors simply prefer intermediary assets over outside assets on average. Indeed, to provide liquidity insurance without the risk of runs, financial institutions and benevolent regulators have an incentive to lower costs for intermediaries transferring assets to their own depositors while raising costs for institutions selling assets to depositors of other institutions. In the language of Proposition 2, this means lowering $\varepsilon_H$ and $\varepsilon_L$ relative to $\varepsilon$ to increase the degree of insurance that is strongly implementable. One way for intermediaries to do so is to set up contracts in advance that specify how asset transfers will occur in a run. On the other hand, regulation can directly limit asset sales from intermediaries to depositors at other intermediaries.

## 3.2 Comparison to Suspension

There is a long history of researchers and policy makers proposing or implementing suspension mechanisms to try to eliminate bank runs. These policies have worked better in theory than in practice, and have frequently been controversial due to concerns over their legality, fairness, feasibility, and sub-optimality. We focus on the latter two, and formalize reasons why the asset transfer

mechanism is preferable to suspension. Specifically, we consider three extensions to our baseline environment: limited commitment, repeat visits to the intermediary during period 1, and aggregate return risk. In the first two, suspension may fail to stop self-fulfilling runs because it does not alleviate the concerns of anxious patient depositors, whereas the asset transfer mechanism remains effective. In the final extension, a run may be inevitable, in which case suspension makes the problem even worse, whereas asset transfers have no effect.

We focus on allocations of the same form as the first best: impatient depositors consume $\bar{c}_1$, patient depositors consume $\bar{c}_2$, the allocation is feasible, which means it satisfies the resource constraint $(1-\lambda)\bar{c}_2 = R(1-\lambda\bar{c}_1)$, and the allocation provides insurance, which means $1 < \bar{c}_1 < \bar{c}_2 < R$. A suspension mechanism offers depositors the options to withdraw or wait in period 1, as in the demand deposit mechanism, but only allows a maximum measure $\gamma$ of depositors to withdraw. After $\gamma$ withdrawals, all remaining depositors in period 1 must wait. We suppose that if an impatient depositor shows up in period 1 after withdrawals are suspended, then they come back in period 2 to receive their share of the return on the intermediary's assets.[15] We impose that $\gamma \in [\lambda, (R/\bar{c}_1 - 1)/(R-1))$. The lower bound implies that suspension weakly implements $(\bar{c}_1, \bar{c}_2)$ in our baseline setup by allowing all impatient depositors to withdraw in period 1. The upper bound implies that suspension strongly implements $(\bar{c}_1, \bar{c}_2)$ in our baseline setup because even if all depositors attempt to withdraw in period 1, a depositor who waits will receive strictly more than $\bar{c}_1$ in period 2. Finally, for simplicity, we suppose asset transfers are frictionless, i.e., $R^a = R$ or equivalently, $\varepsilon = 0$.

### 3.2.1 Commitment

As highlighted by Ennis and Keister (2009, 2010), the suspension mechanism faces a potential commitment problem: if there is a run and suspension is imposed, then a benevolent intermediary or government wants to lift the suspension and dispense goods to remaining impatient depositors. This may leave little enough left for remaining patient depositors that it was indeed optimal to

[15] We can justify this as supposing impatient depositors receive a small amount of utility from period 2 consumption.

run in the first place. In this section, we formalize the commitment issue in a way that allows for a comparison of the suspension and asset transfer mechanisms. We show that the asset transfer mechanism is not subject to the same criticism.

Consider the baseline environment from Section 2.1 but now allow for limited commitment. Suppose after a measure $\lambda$ of depositors interact with the intermediary in period 1 (the earliest the intermediary would suspend withdrawals), the intermediary sets a new mechanism to maximize the expected utility of its depositors. When the intermediary sets its initial mechanism, it is common knowledge that the intermediary cannot commit to the mechanism it will choose in the middle of period 1. The equilibrium concept is subgame perfect Nash: given the initial mechanism, agents act optimally based on accurate beliefs about the mechanism that the intermediary will optimally choose in the middle of period 1 and the equilibrium that mechanism will induce.

**Proposition 3.** *As in Ennis and Keister (2009), if depositor utility is given by* $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ *and the coefficient of relative risk aversion,* $\gamma$*, is sufficiently large, then the suspension mechanism cannot strongly implement the first best allocation. On the other hand, the asset transfer mechanism can always strongly implement the first best.*

*Proof.* See Appendix C. □

The intermediary has difficulty committing to suspension because if a run occurs, then when it is time to suspend, there are impatient depositors left to serve. Moreover, since running patient depositors depleted the intermediary's resources more than usual, the intermediary wants to reduce payouts to remaining depositors, both impatient and patient. Thus, patient depositors want to run and take goods before the suspension if possible. The asset transfer mechanism does not face the same commitment problem for two reasons. First, impatient depositors are never turned away, so the intermediary does not feel the same temptation to abandon protocol. Second, patient depositors are never tempted to take goods early, and instead take the asset transfer if they fear a run. Since the asset transfer depletes the intermediary's resources *less* than usual, if a run pushes the intermediary to change its mechanism, it wants to *increase* payouts to remaining depositors.

### 3.2.2 Repeat Visits

The classic Diamond and Dybvig (1983) setup imposes the stark assumption that a depositor knows their lifetime liquidity needs when they first visit the intermediary. Engineer (1989) demonstrates that if we relax this assumption, then suspension may no longer stop runs. Intuitively, when depositors first visit, they *all* fear they may be impatient, and so want to withdraw before suspension is imposed. In this section, we consider an extension of our baseline setup that resembles the model in Engineer (1989). We show that asset transfers are more effective than suspension.

Consider our baseline environment from Section 2.1 but now allow for repeat visits during period 1. Each depositor draws two independent places in line, $s_1 \in [0, 0.5]$ and $s_2 \in (0.5, 1]$. At the beginning of the interval $[0, 0.5]$, a measure $\zeta\lambda$ of depositors learn they are impatient, where $\zeta \in [0, 1]$. At the beginning of the interval $(0.5, 1]$, of the remaining $1 - \zeta\lambda$ depositors who do not yet know whether they are impatient, a measure $(1 - \zeta)\lambda$ learn they are impatient, and the other $1 - \lambda$ depositors are patient. Thus, a depositor who does not learn they are impatient at the beginning of the first interval believes they will turn out to be impatient with probability $(1 - \zeta)\lambda/(1 - \zeta\lambda)$. As before, a measure $\lambda$ of depositors end up being impatient.

This extension captures the idea that depositors may have an opportunity to interact with the intermediary before they know whether they are impatient. $\zeta$ indexes the amount of information depositors have when they first visit the intermediary. In one extreme, if $\zeta = 1$, then we are in the baseline model except in period 1, each depositor interacts with the intermediary twice after learning their type. In the other extreme, if $\zeta = 0$, then in period 1, each depositor interacts with the intermediary once before they learn their type, and once after.

We maintain that mechanisms are as in our baseline environment: suspension is characterized by a single threshold of withdrawals $\gamma$ after which no more depositors can withdraw in period 1, and the goods withdrawal and asset transfer size ($\bar{c}_1$ and $\kappa$) are fixed throughout period 1.[16] The

[16] This contrasts with Engineer (1989) in which withdrawals can be suspended after one threshold is breached during the first round of visits, then allowed at the beginning of the second round of visits until another threshold is breached. If we allowed for these multiple suspension thresholds, then it would also make sense to allow for the size of the goods withdrawal (as in Engineer (1989)) and of the asset transfer to vary during the second round of visits as a function of depositor actions during the first round. For the sake of simplicity, we do not allow this level of generality.

following proposition states that the asset transfer mechanism dominates suspension in the sense that it strongly implements for a wider range of $\zeta$.

**Proposition 4.** *Suppose an allocation $(\bar{c}_1, \bar{c}_2)$ is feasible and provides insurance. There exist $\zeta_A \leq \zeta_S$ such that the suspension mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if $\zeta > \zeta_S$, and the asset transfer mechanism can strongly implement if and only if $\zeta > \zeta_A$. Moreover, $\zeta_A < 1$ and if $\bar{c}_2/\bar{c}_1$ is sufficiently low, then $\zeta_S > 0$ and $\zeta_A < \zeta_S$, i.e., there is a non-empty interval of $\zeta$ for which asset transfers can strongly implement $(\bar{c}_1, \bar{c}_2)$ but suspension cannot.*

*Proof.* See Appendix C. □

The proposition first states that if suspension can strongly implement an allocation $(\bar{c}_1, \bar{c}_2)$, then so can the asset transfer mechanism. Second, if the allocation offers sufficiently high insurance ($\bar{c}_2/\bar{c}_1$ is sufficiently low), then there is an interval $(\zeta_A, \zeta_S]$ of values for the information parameter $\zeta$ such that the asset transfer mechanism can strongly implement the allocation $(\bar{c}_1, \bar{c}_2)$ but suspension cannot. Intuitively, if a depositor takes the asset transfer during their first visit and turns out to be impatient, then they can liquidate it for positive value. On the other hand, if a depositor waits during their first visit and turns out to be impatient after suspension is triggered, then they receive nothing. Suspension thus ends up looking like asset transfers that cannot be liquidated by depositors in period 1. As a result, asset transfers are more effective.

### 3.2.3 Aggregate Return Risk

So far, we have studied environments in which the efficient way for the intermediary to provide goods to patient depositors is to hold assets to maturity. We now allow for the possibility that it is efficient to liquidate assets because of a fall in the return. Suspension then leads to a worse outcome than the demand deposit and asset transfer mechanisms because it forces assets to remain on the intermediary's balance sheet.

One interpretation is that the intermediary is not benevolent and wants to survive until period 2. It uses suspension to stop depositors from forcing the liquidation of its assets. This is in the spirit

of a number of papers that demonstrate that giving depositors the right to force liquidation may be a powerful disciplining device on intermediary behavior (for example, Calomiris and Kahn (1991) and Diamond and Rajan (2001)). We show that the asset transfer mechanism prevents runs, and unlike suspension, still allows depositors to play this disciplining role. Along with the previous discussion of commitment, this shows that asset transfers perform better than suspension whether or not the intermediary is benevolent.

Consider the baseline environment from Section 2.1 but now allow for aggregate return risk. Suppose that with probability $\theta \in (0,1)$, the long-term return on the asset is $R$, and with probability $1-\theta$, the return is 0. Regardless of the return, the asset can be liquidated in period 1 for 1 good, as before. The value of the long-term return is realized at the beginning of period 1 and is common knowledge. We restrict attention to mechanisms in which the options offered do not depend on the realization of the long-term return.[17]

**Proposition 5.** *Suppose $(\bar{c}_1, \bar{c}_2)$ is feasible and provides insurance when the long-term return is R. Set the asset transfer to be $\kappa = \bar{c}_2/R$ in the asset transfer mechanism. When the long-term return is R, both the asset transfer and suspension mechanisms strongly implement the first best. When the long-term return is 0, the unique equilibrium under the asset transfer mechanism has strictly higher welfare than the unique equilibrium under the suspension mechanism.*

*Proof.* See Appendix C. □

Suspension is worse than the demand deposit mechanism because it prevents the forced, yet efficient, liquidation of the intermediary's assets; patient depositors run, but suspension kicks in and closes the intermediary until period 2. By contrast, the asset transfer mechanism is equivalent to the demand deposit mechanism; patient depositors withdraw goods early, and force the intermediary to liquidate its assets.

[17] This captures that the intermediary cannot contract on the long-term return, and that giving the intermediary discretion does not work—possibly because the intermediary and depositors have a conflict of interest over how to respond if the return is 0.

## 3.3 Idiosyncratic Return Risk

A clear economic implication of our results thus far is that we need to understand the impediments to transferring assets from intermediaries to depositors in order to understand why runs can occur in a Diamond and Dybvig (1983) environment. In this section, we suppose the intermediary holds a diversified portfolio with a deterministic return, but can only transfer units of the asset that carry unobservable idiosyncratic risk. We show that if this risk is sufficiently large, then asset transfers are no longer effective because an asset transfer that patient depositors prefer over the risk free withdrawal of goods is too expensive. Compared to the asset transfer cost $\varepsilon$ in our baseline model, this provides a more direct motivation for assets to remain with the intermediary rather than with depositors between periods 1 and 2, even though the intermediary's only ability is aggregation. One interpretation of our result is that an increase in idiosyncratic return risk across an intermediary's assets can make the intermediary run prone even if there is no change in the average return. Indeed, Ospina and Uhlig (2018) argue that during the 2007-09 financial crisis, the increase in the *dispersion* of loss rates across mortgage backed securities was relatively high compared to the increase in the *average* loss rate.

### 3.3.1 Environment Changes

Consider the baseline environment from Section 2.1, but replace the investment technology with the following. In period 0, an agent chooses a number of investments, and the size of each. An investment of $x$ units of the endowment yields an asset that generates $x$ units of the good if interrupted in period 1 and $Rx$ units of the good if held until period 2, where $R$ is randomly drawn at the beginning of period 2, is i.i.d. across investments:

$$R = \begin{cases} R_H & \text{with probability } q \\ R_L & \text{with probability } 1-q \end{cases}$$

and the expected return is $\bar{R} = qR_H + (1-q)R_L$. We impose that each investment must be of size greater than or equal to $\bar{x} > 0$. This stylized restriction captures a decreasing relationship between the size of an investment and its average cost, perhaps due to a fixed cost of starting a new investment.[18] It implies that each depositor can only make finitely many investments, leaving them exposed to not only liquidity preference risk (as before) but also idiosyncratic return risk. An intermediary, by contrast, can make uncountably many investments, and so will create a fully diversified portfolio with non-random period 2 return $\bar{R}$.

We further impose that each asset and the ownership of each asset are perfectly indivisible; all the intermediary can do with an asset is liquidate, transfer, or hold it until period 2 *in its entirety*. In particular, the intermediary cannot create a new, diversified, and transferable asset by mixing together fractions of assets. Thus, even though the intermediary fully diversifies its portfolio, it can only transfer assets that bear the same idiosyncratic risks a depositor would have faced had they invested on their own. We interpret this as allowing the intermediary to provide insurance by pooling resources but limiting its ability to securitize its portfolio.[19]

Finally, for simplicity, we suppose asset transfers are frictionless, i.e., $\varepsilon = 0$. For technical reasons, we restrict preferences so that $\lim_{c \to 0} u(c) = -\infty$.

### 3.3.2 Intermediary Problem

We now characterize when the asset transfer mechanism can strongly implement the first best allocation in this setting. The constrained planner problem and the first best allocation are the same as in Section 2.3 but with $R$ replaced by $\bar{R}$. For simplicity, we suppose parameters are such that it is sufficient to consider the case in which the intermediary offers, as its asset transfer, one asset of size $\kappa$ to each depositor and, when choosing $\kappa$, the constraint $\kappa \geq \bar{x}$ does not bind.[20]

[18] The minimum size makes our investment technology similar to the one in Diamond (1984) and Williamson (1986), although for them, it plays a different role of justifying costly monitoring.

[19] We consider the case in which the intermediary effectively cannot perform any securitization, but we believe the results are qualitatively unchanged as long as the intermediary cannot perfectly securitize its portfolio.

[20] Specifically, we suppose parameters are such that 1) if the intermediary can strongly implement the first best by offering, as its asset transfer, a single asset of size $\kappa$, then $\kappa \geq \bar{x}$; and 2) if the intermediary cannot strongly implement the first best with any such asset transfer, then $\bar{x}$ is sufficiently large that the intermediary also cannot strongly implement the first best by offering to transfer two or more assets to any depositors.

Hence, the intermediary makes a measure $1/\kappa$ of investments, each of size $\kappa$.

The following proposition states that for fixed preferences and a fixed average return, the asset transfer mechanism can strongly implement the first best allocation if and only if the dispersion of idiosyncratic returns is sufficiently low.

**Proposition 6.** *Vary $\eta \equiv R_H/R_L$, and adjust the return realizations $R_H$ and $R_L$ so that the average return, $\bar{R}$, and the probability of a high return, $q$, are constant. There exists a $\bar{\eta} \in (1,\infty)$ such that the asset transfer mechanism can strongly implement the first best allocation if and only if $\eta < \bar{\eta}$.*

*Proof.* See Appendix D. □

A patient depositor that takes the asset transfer now faces idiosyncratic return risk. Hence, if return dispersion $\eta$ is higher, then the intermediary must offer a higher transfer size $\kappa$ to maintain patient depositors' preference for the risky asset transfer over the risk free early withdrawal of goods. As $\eta$ increases, the required $\kappa$ is eventually so large that a new type of run equilibrium emerges in which all patient depositors take the asset transfer, and the intermediary runs out of resources before the end of period 1. In this sense, idiosyncratic return risk plays the same role as the asset transfer cost $\varepsilon$.

Finally, as with the transfer cost in Corollary 1, even if $\eta \geq \bar{\eta}$, the asset transfer mechanism may still be able to strongly implement some insurance, unlike the demand deposit mechanism.

*Additional considerations:* In this subsection, we focused on how idiosyncratic return risk affects risk averse depositors' value of accepting asset transfers. There are additional issues that may arise in an environment with heterogeneous assets. For example, suppose depositors can identify asset quality before choosing which assets they can get in a transfer, or before choosing whether to accept an offered asset transfer. This may lead to a type of run in which early depositors take all the good assets, leaving only bad assets for late depositors. There are many ways to model such an environment. We explore one possibility formally in Online Appendix E in which depositors are offered a single asset transfer, and they can see whether it is high or low quality before choosing

whether to accept. We find that strong implementation of the first best is *not* possible if and only if return dispersion $\eta$ is sufficiently large and the probability $q$ of a high return is sufficiently intermediate. Intuitively, if most assets are low return ($q$ is low), then the low return $R_L$ must be close to the average return $\bar{R}$, so depositors are not worried about getting stuck with low return assets. On the other hand, if most assets are high return ($q$ is high), then the intermediary can afford early withdrawals by the small share of patient depositors that are offered low return assets.

# 4 Conclusion

We showed that a benevolent intermediary facing the classic Diamond and Dybvig (1983) environment can choose a simple mechanism that strongly implements the first best allocation. This mechanism involves offering asset transfers that are sufficiently large to prevent patient depositors from withdrawing goods early but sufficiently small that the intermediary never runs out of resources from transferring assets. We interpret this mechanism as a repo contract in which the intermediary (borrower) gives the depositor (lender) an asset (collateral) that can be kept if the depositor is worried about a run, as a covered bond, or as a bankruptcy plan that gives depositors the option to progressively disassemble an insolvent intermediary. We also showed that this mechanism dominates suspension if policy makers have limited commitment, if depositors repeatedly interact with the intermediary, and/or if there is aggregate return risk.

The spirit of our mechanism is that financial intermediaries are not the only potential holders of intermediary assets, especially during a run. Thus, it may be valuable to facilitate the transfer of assets from institutions to their depositors in case of a run. At the same time, to prevent the unraveling of liquidity insurance, it is important to maintain frictions in the trade of assets between financial institutions and depositors *at other institutions*. One way to balance these two interests is to set up contracts in advance that allow financial intermediaries and their depositors to negotiate over who holds intermediary assets when a run occurs.

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# A Proofs for Section 2.1

**Lemma 1.** *Define a mechanism $\Gamma$ that offers depositors three options: "withdraw", "wait", and "transfer", where the first two are as in the demand deposit mechanism, and if a depositor chooses "transfer", then they only receive units of the asset.*

*If there does not exist a constant $\kappa$ such that the asset transfer mechanism strongly implements the first best, then $\Gamma$ does not strongly implement the first best.*

*Proof.* Suppose the asset transfer mechanism cannot strongly implement the first best allocation. It follows from Theorem 1 that $R^a \leq \frac{R}{c_2^*/c_1^*}$. We now show that under the mechanism $\Gamma$, a suboptimal equilibrium exists. Suppose impatient depositors "withdraw" and patient depositors choose whichever they prefer of "withdraw" and "transfer", breaking ties in favor of the former. In this equilibrium, if no depositors are offered an asset transfer $\kappa > c_1^*/R^a$, then a traditional run equilibrium exists, and we are done. Suppose instead that a fraction $\beta > 0$ of depositors are offered an asset transfer $\kappa > c_1^*/R^a$, which they take if patient. Define $\bar{\kappa}$ to be the average value of $\kappa$ offered to such depositors. It follows that $\bar{\kappa} > c_1^*/R^a \geq c_2^*/R$. Hence, the intermediary runs out of resources strictly before the end of period 1 because if not, its resources remaining at the end of period 1 would be

$$1 - \lambda c_1^* - (1-\lambda)(1-\beta)c_1^* - (1-\lambda)\beta\bar{\kappa},$$

which is strictly less than 0 since $\lambda c_1^* + (1-\lambda)c_2^*/R = 1$ and $c_1^* > c_2^*/R$. As such, a depositor who waits would receive nothing in period 2, so the specified strategy profile is indeed optimal. The equilibrium does not achieve the first best allocation since a positive measure of depositors receive no goods in either period. □

*Proof of Theorem 1.* Necessity: We show the contrapositive: if $R^a \leq \frac{R}{c_2^*/c_1^*}$, then the asset transfer mechanism cannot strongly implement the first best allocation, i.e., a suboptimal equilibrium exists. Suppose $R^a \leq \frac{R}{c_2^*/c_1^*}$. We consider three cases. First, suppose $\kappa \leq c_1^*/R^a$. Then, both impatient and patient depositors weakly prefer to withdraw goods in period 1 over taking the asset transfer. Hence, the traditional run equilibrium in which all depositors withdraw goods in period 1 exists.

Next, suppose $\kappa \geq c_1^*$. We show that there is a suboptimal equilibrium in which all depositors take the asset transfer. Both impatient and patient depositors weakly prefer to take the asset transfer over withdrawing goods in period 1. In the proposed equilibrium, the intermediary runs out of assets strictly before the end of period 1 because serving all depositors requires a measure $\kappa \geq c_1^* > 1$ assets. Thus, the proposed equilibrium is an equilibrium because as in a traditional run, any depositor who waits receives 0 goods. Moreover, the equilibrium is suboptimal because as in a traditional run, a positive measure of depositors receive 0 goods.

Finally, suppose $c_1^*/R^a < \kappa < c_1^*$. We show that there is a suboptimal equilibrium in which impatient depositors withdraw goods in period 1, and patient depositors take the asset transfer. Impatient depositors strictly prefer to withdraw goods in period 1 over taking the asset transfer, and patient depositors strictly prefer to take the asset transfer over withdrawing goods in period 1. Moreover, since $R^a \leq \frac{R}{c_2^*/c_1^*}$, it follows that $\kappa > c_2^*/R$. As such, in the proposed equilibrium, the intermediary runs out of resources strictly before the end of period 1 because the transfer requires more assets than must be used to supply a depositor with $c_2^*$ goods in period 2. Formally, serving

all depositors requires a measure of assets equal to

$$\lambda c_1^* + (1-\lambda)\kappa > \lambda c_1^* + (1-\lambda)c_2^*/R = 1.$$

Thus, the proposed equilibrium is an equilibrium because as in a traditional run, any depositor who waits receives 0 goods. Moreover, the equilibrium is suboptimal because as in a traditional run, a positive measure of depositors receive 0 goods.

Sufficiency: We show that if $R^a > \frac{R}{c_2^*/c_1^*}$, then there exists an asset transfer $\kappa$ such that the asset transfer mechanism strongly implements the first best allocation. Specifically, set $\kappa = c_2^*/R$. Since $\kappa < c_1^*$, impatient depositors strictly prefer to withdraw goods in period 1 over taking the asset transfer. Moreover, they also do not want to wait to withdraw until period 2. Since $\kappa > c_1^*/R^a$, patient depositors strictly prefer to take the asset transfer over withdrawing goods in period 1. It follows that in any equilibrium, impatient depositors withdraw goods in period 1, and patient depositors take the asset transfer with some probability $\beta \in [0,1]$ and otherwise wait. Then, at the end of period 1, the measure of the intermediary's asset holdings remaining is

$$\begin{aligned} B(1;\mathbf{a}) &= 1 - \lambda c_1^* - (1-\lambda)\beta\kappa \\ &= 1 - \lambda c_1^* - (1-\lambda)\beta\frac{c_2^*}{R} \\ &= (1-\lambda)(1-\beta)\frac{c_2^*}{R}, \end{aligned}$$

where the last equality follows from the fact that the resource constraint binds in the first best. Hence, the intermediary gives out a measure $(1-\lambda)(1-\beta)c_2^*$ of goods in period 2 to a measure $(1-\lambda)(1-\beta)$ of waiting depositors, so each depositor who waits receives $c_2^*$ goods.

There are two cases. If $R^a < R$, then patient depositors strictly prefer to wait than to take the asset transfer because the former generates $c_2^*$ goods and the latter generates $R^a\kappa = \frac{R^a}{R}c_2^*$ goods. Thus, there is a unique equilibrium, and it entails impatient depositors withdrawing goods in period 1, and patient depositors waiting to withdraw goods in period 2. This equilibrium yields the first best allocation. On the other hand, if $R^a = R$, then patient depositors are indifferent between taking the asset transfer and waiting to withdraw in period 2. Thus, any degree of mixing $\beta$ is an equilibrium, and they all yield the first best allocation.

As an aside, the above arguments show that if the asset transfer mechanism strongly implements the first best allocation, then the first best is also the unique outcome that survives the iterated deletion of strictly dominated strategies. □

# B Proofs for Section 3.1

Before proving Propositions 1 and 2, we state and prove the following lemma.

**Lemma 2.** *Suppose the intermediary uses the demand deposit or asset transfer mechanism with period 1 goods withdrawal c. If a patient depositor can consume at most x by taking the asset*

*transfer or waiting, then they must withdraw goods early if $R^a c > x$, must not withdraw goods early if $R^a c < x$, and can withdraw goods early with any probability if $R^a c = x$.*

*Proof.* If $R^a c > x$, then the patient depositor must withdraw goods and use them to purchase assets from outside agents because there is a strict mutual benefit to doing so: they can give the $c$ goods to outside agents for $A$ assets where $x/R^a < A < c$; outside agents are strictly better off than if they had liquidated their $A$ assets because $c > A$; the depositor is strictly better off than if they had taken their best alternative option because they hold the $A$ assets to maturity and consume $R^a A > x$. On the other hand, if $R^a c < x$, then the patient depositor must not withdraw goods because even if outside agents give up enough assets to be indifferent between trading and not trading ($c$ assets for $c$ goods), then the patient depositor only consumes $R^a c < x$ by withdrawing goods and trading with outside agents, whereas they consume $x$ by choosing their best alternative. Finally, if $R^a c = x$, then the patient depositor can withdraw goods to trade with outside agents, but also can choose not to because the mutual benefit from doing so is exactly zero. □

We now prove Propositions 1 and 2 together because the latter nests the former.

*Proof of Propositions 1 and 2.* We first prove the results on the demand deposit mechanism and then on the asset transfer mechanism. The demand deposit mechanism weakly implements $(\bar{c}_1, \bar{c}_2)$ if and only if there is an equilibrium in which impatient depositors withdraw goods and patient depositors wait. In such an equilibrium, a depositor who withdraws early gets $\bar{c}_1$ goods and a depositor who waits gets $\bar{c}_2$. Impatient depositors prefer to withdraw goods over waiting because they do not value goods in period 2. By Lemma 2, patient depositors can choose to wait over withdrawing goods if and only if $R^a \bar{c}_1 \leq \bar{c}_2$. Thus, weak implementation is possible if and only if $\bar{c}_2/\bar{c}_1 \geq R^a$. Moreover, the demand deposit mechanism cannot strongly implement any insurance by the same argument as in Diamond and Dybvig (1983).

Next, we show the results for the asset transfer mechanism in both the homogeneous frictions and heterogeneous frictions cases. To do so, we prove the results for the heterogeneous frictions case and note that this nests the homogeneous frictions case by setting $R^a_H = R^a = R^a_L$. We refer to a depositor who earns effective return $R^a_L \leq R^a$ on transferred intermediary assets as an $L$-type depositor and a depositor who earns effective return $R^a_H \geq R^a$ on transferred intermediary assets as an $H$-type depositor. The share of depositors that are $H$-type is $\alpha$.

For the results on weak implementation, we have already shown one direction: the asset transfer mechanism can weakly implement $(\bar{c}_1, \bar{c}_2)$ if $\bar{c}_2/\bar{c}_1 \geq R^a$ because the asset transfer mechanism nests the demand deposit mechanism by setting the asset transfer to $\kappa = 0$. For the other direction, suppose $\bar{c}_2/\bar{c}_1 < R^a$. For impatient depositors to consume at least $\bar{c}_1$, it must be that the goods withdrawal is $c \geq \bar{c}_1$ or the asset transfer is $\kappa \geq \bar{c}_1/(1-\varepsilon)$. If $c \geq \bar{c}_1$, then a patient depositor can get $R^a c \geq R^a \bar{c}_1 > \bar{c}_2$ goods by taking the goods withdrawal and purchasing assets. If $\kappa \geq \bar{c}_1/(1-\varepsilon)$, then some patient depositors can get $R^a_H \kappa \geq R^a \kappa \geq R\bar{c}_1 > \bar{c}_2$ goods by taking the asset transfer and holding it to maturity. Thus, in any equilibrium, some patient depositors get strictly more than $\bar{c}_2$ goods. Since $(\bar{c}_1, \bar{c}_2)$ satisfies the resource constraint, it follows that some impatient depositors get strictly less than $\bar{c}_1$ or some patient depositors get strictly less than $\bar{c}_2$. So there cannot be any equilibrium that achieves allocation $(\bar{c}_1, \bar{c}_2)$ or any allocation that Pareto dominates $(\bar{c}_1, \bar{c}_2)$.

To complete the proof of the propositions, we show that the asset transfer mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if $\bar{c}_2/\bar{c}_1 > (\alpha R^a/R^a_H + 1 - \alpha)R$. Note that this inequality becomes $\bar{c}_2/\bar{c}_1 > R$ in the homogeneous frictions case ($R^a_H = R^a$), which is inconsistent with

$(\bar{c}_1, \bar{c}_2)$ providing liquidity insurance. As such, in the homogeneous frictions case, this statement is equivalent to stating that the asset transfer mechanism cannot strongly implement $(\bar{c}_1, \bar{c}_2)$. We first show the "only if" direction (necessity) and then the "if" direction (sufficiency).

Necessity: We prove the contrapositive that if $(\bar{c}_1, \bar{c}_2)$ is strongly implementable, then $\bar{c}_2/\bar{c}_1 > (\alpha R^a/R^a_H + 1 - \alpha)R$. First, if $(\bar{c}_1, \bar{c}_2)$ is strongly implementable, then it must be weakly implementable, which requires that $\bar{c_2}/\bar{c}_1 \geq R^a$, as we just proved. Moreover, to weakly implement $(\bar{c}_1, \bar{c}_2)$, the goods withdrawal must be $\bar{c}_1$. To see why, observe that weak implementation requires $\kappa < \bar{c}_1/(1-\varepsilon)$ because otherwise, a patient $H$-type depositor can consume $R^a_H \kappa \geq R\bar{c}_1 > \bar{c}_2$ goods by holding the asset transfer to maturity, which means some depositors must consume strictly less than $\bar{c}_1$ (if impatient) or $\bar{c}_2$ (if patient) in any equilibrium. As such, for impatient depositors to be able to consume $\bar{c}_1$ goods, weak implementation requires the goods withdrawal to be $\bar{c}_1$.

To complete the proof of "necessity", we now show various inequalities the asset transfer $\kappa$ must satisfy in order to strongly implement $(\bar{c}_1, \bar{c}_2)$. Since $\bar{c}_1 > 1$, it follows that the asset transfer mechanism can only strongly implement $(\bar{c}_1, \bar{c}_2)$ if the asset transfer satisfies $\kappa < 1$. Otherwise, there is a run equilibrium in which all depositors either withdraw goods early or take the asset transfer. Next, $\kappa < 1$ means impatient depositors always withdraw goods early because they strictly prefer doing so over taking the asset transfer and because they do not value goods in period 2. Moreover, by Lemma 2, $L$-type patient depositors never take the asset transfer because doing so yields $R^a_L \kappa < R^a_L \bar{c}_1 \leq R^a \bar{c}_1$ goods. It follows that the asset transfer mechanism can only strongly implement $(\bar{c}_1, \bar{c}_2)$ if patient $H$-type depositors strictly prefer taking the asset transfer over withdrawing goods early, i.e., if $R^a_H \kappa > R^a \bar{c}_1$. Otherwise, since other depositors prefer withdrawing goods early over taking the asset transfer, there is a classic run equilibrium in which all depositors withdraw goods early.

Finally, for strong implementation to be possible, it must be that if impatient and patient $L$-type depositors withdraw goods early and patient $H$-type depositors take the asset transfer, then the intermediary does not run out of resources by the end of period 1; otherwise, this is an equilibrium that does not implement $(\bar{c}_1, \bar{c}_2)$ because the intermediary runs out of resources before the end of period 1. Thus, it must be that

$$(\lambda + (1-\lambda)(1-\alpha))\bar{c}_1 + (1-\lambda)\alpha\kappa \leq 1,$$

where $\lambda$ is the share of impatient depositors and $\alpha$ is the independent share of $H$-type depositors. Subtracting $\lambda\bar{c}_1$ from both sides and plugging in the resource constraint, $(1-\lambda)\bar{c}_2 = R(1-\lambda\bar{c}_1)$, we can write this inequality as

$$(1-\alpha)\bar{c}_1 + \alpha\kappa \leq \bar{c}_2/R.$$

This can be consistent with $R^a_H \kappa > R^a \bar{c}_1$—which we showed is required for strong implementation—if and only if

$$\bar{c}_2/\bar{c}_1 > (\alpha R^a/R^a_H + 1 - \alpha)R.$$

This completes the proof of "necessity".

Observe that with homogeneous frictions, $R^a = R^a_H$, so our requirement for strong implementation becomes $\bar{c}_2/\bar{c}_1 > R$. This does not hold for any allocation $(\bar{c}_1, \bar{c}_2)$ that provides liquidity insurance, so strong implementation is never possible, as stated in Proposition 1.

Sufficiency: We suppose $\bar{c}_2/\bar{c}_1 > (\alpha R^a/R^a_H + 1 - \alpha)R$ and show that the asset transfer mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$. Immediately, this means we are in the heterogeneous frictions case where there is a strictly positive share of $H$-type depositors ($\alpha > 0$) and they earn a strictly higher return on transferred intermediary assets than on purchased outside assets ($R^a_H > R^a$).

Set the goods withdrawal to $\bar{c}_1$. From the "necessity" proof, we know we can set an asset transfer $\kappa$ such that $H$-type patient depositors strictly prefer the asset transfer over withdrawing goods early—$R^a_H \kappa > R^a \bar{c}_1$—and such that the intermediary can strictly afford early goods withdrawals by all impatient and patient $L$-type depositors and asset transfers for all patient $H$-type depositors—$(1-\alpha)\bar{c}_1 + \alpha\kappa < \bar{c}_2/R$. It follows from the first inequality that patient $H$-type depositors mix between taking the asset transfer and waiting; let $\beta_H$ be the probability they do the former. It follows from the second inequality that $\kappa < \bar{c}_2/R < \bar{c}_1$, so impatient depositors always withdraw goods early and patient $L$-type depositors mix between withdrawing goods early and waiting; let $\beta_L$ be the probability they do the former. It further follows from the second inequality that for any values of $\beta_L$ and $\beta_H$, the intermediary has a strictly positive measure of assets remaining at the end of period 1.

To complete the proof, we show that for any values of $\beta_L$ and $\beta_H$ (the probabilities patient $L$-type and patient $H$-type depositors run, respectively), all patient depositors strictly prefer to wait. This then means there is a unique equilibrium, which is the truth-telling equilibrium in which impatient depositors withdraw goods early and patient depositors wait. For any value of $\beta_L$ and $\beta_H$, the intermediary has a strictly positive measure of assets remaining at the end of period 1, so a depositor who waits gets strictly positive period 2 consumption

$$c_2(\beta_L, \beta_H) = R\frac{1 - \lambda\bar{c}_1 - (1-\lambda)(1-\alpha)\beta_L\bar{c}_1 - (1-\lambda)\alpha\beta_H\kappa}{(1-\lambda)((1-\alpha)(1-\beta_L) + (1-\lambda)\alpha(1-\beta_H))}.$$

We now show that $c_2(\beta_L, \beta_H) > R\kappa$ for all $\beta_L$ and $\beta_H$; we then show that this implies all patient depositors strictly prefer to wait. Note that $c_2(\beta_L, \beta_H)$ goes to positive infinity as $\beta_L$ and $\beta_H$ go to 1. Take the partial derivatives of $c_2(\beta_L, \beta_H)$ to get

$$\frac{\partial c_2(\beta_L, \beta_H)}{\partial \beta_L} = \frac{(1-\lambda)(1-\alpha)}{(1-\lambda)((1-\alpha)(1-\beta_L) + (1-\lambda)\alpha(1-\beta_H))}(c_2(\beta_L, \beta_H) - R\bar{c}_1)$$

and

$$\frac{\partial c_2(\beta_L, \beta_H)}{\partial \beta_H} = \frac{(1-\lambda)\alpha}{(1-\lambda)((1-\alpha)(1-\beta_L) + (1-\lambda)\alpha(1-\beta_H))}(c_2(\beta_L, \beta_H) - R\kappa).$$

By way of contradiction, suppose $c_2(\beta^*_L, \beta^*_H) \leq R\kappa$ for some $\beta^*_L$ and $\beta^*_H$. From the partial derivative with respect to $\beta_H$, it follows that $c_2(\beta^*_L, 1) \leq R\kappa$. Then since $\kappa < \bar{c}_1$, it follows from the partial derivative with respect to $\beta_L$ that $c_2(\beta_L, 1) < R\kappa$ for all $\beta_L > \beta^*_L$. This contradicts that $c_2(\beta_L, \beta_H)$ goes to positive infinity as $\beta_L$ and $\beta_H$ go to 1, so it must be that $c_2(\beta_L, \beta_H) > R\kappa$ for all $\beta_L$ and $\beta_H$.

Finally, in any equilibrium, patient $H$-type depositors strictly prefer to wait over taking the asset transfer because the latter yields $R^a_H \kappa \leq R\kappa < c_2(\beta_L, \beta_H)$ goods. Patient $L$-type depositors strictly prefer to wait over withdrawing goods early because the latter yields at most $R^a\bar{c}_1 < R^a_H\kappa < c_2(\beta_L, \beta_H)$ goods, where the first inequality was by construction at the beginning of the "sufficiency" proof. Thus, there is an unique equilibrium: impatient depositors withdraw goods early

and patient depositors wait. Hence, the asset transfer mechanism strongly implements $(\bar{c}_1, \bar{c}_2)$, which completes the proof of "sufficiency". □

# C Proofs for Section 3.2

*Proof of Proposition 3.* For the suspension mechanism, the proof is as in Ennis and Keister (2009, 2010). Note that the proof is the same for any value of the suspension threshold above its minimum, i.e., $\gamma \geq \lambda$ because after a measure $\lambda$ of depositors, the intermediary chooses a new mechanism.

Consider the asset transfer mechanism. Suppose the asset transfer is $\kappa = c_2^*/R$ at first. Regardless of the mechanism that the intermediary will choose after interacting with a measure $\lambda$ of depositors, a patient depositor strictly prefers to take the asset transfer rather than withdraw goods in period 1; each yields a value independent of the intermediary and other depositors, and the asset transfer yields a strictly higher value for patient depositors. On the other hand, an impatient depositor always strictly prefers to withdraw goods rather than take the asset transfer or wait until period 2. Suppose a fraction $\alpha$ of patient depositors in the first $\lambda$ measure of depositors take the asset transfer, and the rest wait until period 2. Then, when the intermediary chooses their new mechanism, the measure of assets remaining on their balance sheet is

$$1 - \lambda^2 c_1^* - \lambda(1-\lambda)\alpha c_2^*/R.$$

A measure $1-\lambda$ of depositors remain to interact with the intermediary in period 1, a fraction $\lambda$ of whom are impatient, and a measure $\lambda(1-\lambda)(1-\alpha)$ of patient depositors have already decided to wait until period 2 to withdraw. Thus, the new first best allocation consists of a $c_1^{**}$ and $c_2^{**}$ that satisfy the first order condition:

$$u'(c_1^{**}) = Ru'(c_2^{**}),$$

and the budget constraint:

$$R[1 - \lambda^2 c_1^* - \lambda(1-\lambda)\alpha c_2^*/R - (1-\lambda)\lambda c_1^{**}] = [\lambda(1-\lambda)(1-\alpha) + (1-\lambda)^2]c_2^{**}.$$

We can see that $c_1^{**} = c_1^*$ and $c_2^{**} = c_2^*$ because patient depositors who took the asset transfer used up as many units of the asset as they would had they waited and received $c_2^*$, leaving the intermediary in effectively the same position as at the beginning of period 1. It follows that the intermediary strongly implements the new first best, which is the same as the old first best, with the asset transfer mechanism. Thus, a patient depositor in the initial measure $\lambda$ of depositors is indifferent between taking the asset transfer and waiting, either of which yields $c_2^*$ goods. It follows that the asset transfer mechanism strongly implements the first best. □

*Proof of Proposition 4.* Set the asset transfer to be $\kappa \in (\bar{c}_1/R, \bar{c}_2/R)$ so that patient depositors strictly prefer taking the transfer over withdrawing goods and so that the transfer requires strictly less resources than providing a depositor with $\bar{c}_2$ goods in period 2. As in the baseline model, a depositor who knows they are impatient always chooses to withdraw goods in period 1, which we take as given throughout the proof. We first show there is an equilibrium that yields the allocation $(\bar{c}_1, \bar{c}_2)$, and then turn to whether there is also a run equilibrium.

Suppose depositors believe other depositors are not running under either suspension or the asset transfer mechanism. The decision problem for a patient depositor visiting the intermediary for the

second time in period 1 is the same as in the baseline model, so they wait. A depositor visiting for the first time who does not know whether they are impatient faces the payoffs:

$$\begin{aligned}\mathbb{E}[u(\text{withdraw})] &= u(\bar{c}_1)\\ \mathbb{E}[u(\text{wait})] &= \frac{(1-\zeta)\lambda}{1-\zeta\lambda}u(\bar{c}_1)+\frac{1-\lambda}{1-\zeta\lambda}u(\bar{c}_2)\\ \mathbb{E}[u(\text{transfer})] &= \frac{(1-\zeta)\lambda}{1-\zeta\lambda}u(\kappa)+\frac{1-\lambda}{1-\zeta\lambda}u(R\kappa),\end{aligned}$$

where the last option is only available in the asset transfer mechanism. The weights in the latter two options reflect the probability of ending up impatient vs. patient, and the arguments of $u$ in the last option reflect that a depositor who takes the asset transfer will liquidate in period 1 if they turn out to be impatient and will hold until maturity if they turn out to be patient. Such a depositor prefers waiting over withdrawing because $\bar{c}_2 > \bar{c}_1$. They also prefer waiting over taking the transfer because $\kappa < \bar{c}_2/R < \bar{c}_1$. Thus, there is an equilibrium in which depositors wait if they do not know their type when they first visit or if they know they are patient, and withdraw if they know they are impatient. The equilibrium allocation is $(\bar{c}_1, \bar{c}_2)$.

Now, the only possible run is one in which depositors withdraw goods when visiting for the first time if they do not know whether they are impatient. If these depositors wait, then we are effectively in our baseline environment, so the suspension and asset transfer mechanisms strongly implement $(\bar{c}_1, \bar{c}_2)$. Moreover, if these depositors take the asset transfer, then they use less assets than if they were to withdraw $\bar{c}_1$ or if they were to wait and get $\bar{c}_2$, so as in the proof of Theorem 1, this raises the value of waiting and cannot form an equilibrium.

Thus, under the suspension mechanism, suppose all depositors withdraw goods when they first visit the intermediary.[21] In this case, withdrawals are suspended after a fraction $\gamma < 1$ of depositors first visit, and so before any depositors visit for the second time in period 1. A depositor visiting for the first time who does not know whether they are impatient faces the payoffs:

$$\begin{aligned}\mathbb{E}[u(\text{withdraw})] &= u(\bar{c}_1)\\ \mathbb{E}[u(\text{wait})] &= \frac{(1-\zeta)\lambda}{1-\zeta\lambda}u(0)+\frac{1-\lambda}{1-\zeta\lambda}u(\tilde{c}_2),\end{aligned}$$

where $\tilde{c}_2 = R(1-\gamma\bar{c}_1)/(1-\gamma) \leq \bar{c}_2$ is what each waiting depositor receives in period 2. The second line reflects that if the depositor turns out to be impatient, then they will be unable to withdraw during their second visit in period 1, so they will get utility $u(0)$. The run is an equilibrium if and only if the value of withdrawing is weakly greater than the value of waiting. The latter is decreasing in the suspension threshold $\gamma$ (through $\tilde{c}_2$), so the suspension mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if the value of withdrawing is strictly less than the value of waiting when $\gamma$ is at its minimum value of $\lambda$. In that case, $\tilde{c}_2 = \bar{c}_2$, so the value of waiting is an average of

[21] There is also the possibility of a partial run equilibrium in which only some depositors run when they first visit the intermediary if they are unsure whether they are impatient. However, such a partial run equilibrium can only exist if the full run equilibrium exists because the more depositors who run, the more each depositor wants to run. As such, it is sufficient to focus on full runs.

$u(0)$ and $u(\bar{c}_2)$, where the weight on the former is strictly decreasing in $\zeta$. Therefore, suspension can strongly implement $(\bar{c}_1, \bar{c}_2)$, if and only if $\zeta > \zeta_S$ for some $\zeta_S$. Moreover, as $\bar{c}_2/\bar{c}_1$ decreases to its minimum of 1, $\zeta_S$ must eventually be strictly positive because $u(\bar{c}_2)$ goes to $u(\bar{c}_1)$, and at $\zeta = 0$, the weight on $u(0)$ in the value of waiting is still strictly positive.

Next, under the asset transfer mechanism, suppose all depositors withdraw goods when they first visit the intermediary.[22] Then, the intermediary runs out of resources after a fraction $1/\bar{c}_1 < 1$ of depositors first visit, and so before any depositors visit for the second time in period 1. A depositor visiting for the first time who does not know whether they are impatient faces the payoffs:

$$\begin{aligned}
\mathbb{E}[u(\text{withdraw})] &= u(\bar{c}_1) \\
\mathbb{E}[u(\text{wait})] &= u(0) \\
\mathbb{E}[u(\text{transfer})] &= \frac{(1-\zeta)\lambda}{1-\zeta\lambda} u(\kappa) + \frac{1-\lambda}{1-\zeta\lambda} u(R\kappa),
\end{aligned}$$

where the second line reflects that the intermediary runs out of resources before the depositor's next visit, and the third line reflects that the depositor will liquidate any asset holdings if impatient and hold them to maturity if patient. The run is an equilibrium if and only if the value of taking the transfer is weakly less than the value of withdrawing. Thus, the asset transfer mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if the value of taking the transfer is strictly greater than the value of withdrawing when $\kappa$ is at its supremum value of $\bar{c}_2/R$. At $\kappa = \bar{c}_2/R$, the value of taking the transfer is an average of $u(\bar{c}_2/R)$ and $u(\bar{c}_2)$, where the former is less than the value of withdrawing and the latter is more than the value of withdrawing. Furthermore, the weight on the latter is increasing in $\zeta$ and goes to 1 as $\zeta$ goes to 1. Hence, the asset transfer mechanism can strongly implement $(\bar{c}_1, \bar{c}_2)$ if and only if $\zeta > \zeta_A$ for some $\zeta_A < 1$. Moreover, if the $\zeta$ threshold for suspension, $\zeta_S$, is strictly greater than 0, then it must be that $\zeta_A < \zeta_S$—there is a range of $\zeta$ for which asset transfers strongly implement but suspension does not—because the expected utility from withdrawing is the same in both cases, but the expected utility from a transfer is strictly greater than the expected utility from waiting under suspension. □

*Proof of Proposition 5.* If the return realization is $R$, then suspension works as in Diamond and Dybvig (1983), and the asset transfer mechanism works as in Theorem 1. Now suppose the long-term return is 0. In the suspension mechanism, there is a unique equilibrium: all depositors withdraw goods in period 1. A fraction $\gamma$ (the suspension threshold) of depositors each receive $\bar{c}_1$ in period 1, and the remaining depositors each receive 0. In the asset transfer mechanism, there is a unique equilibrium: all depositors withdraw goods in period 1. A fraction $1/\bar{c}_1$ of depositors each receive $\bar{c}_1$ in period 1, and the remaining depositors each receive 0. Since the ability of suspension to strongly implement when the return realization is $R$ implies that $1/\bar{c}_1 > \gamma$, this allocation yields higher welfare than the allocation in the unique equilibrium under suspension. □

[22] There is also the possibility of a partial run equilibrium. See the previous footnote 21.

# D Proofs for Section 3.3

*Proof of Proposition 6.* Note that as we vary the return dispersion $\eta$, the first best allocation $(c_1^*, c_2^*)$ is constant because we adjust $R_H$ and $R_L$ to keep the average return $\bar{R}$ fixed.

Rearranging the inequality in Theorem 1 shows that in our baseline model, the asset transfer mechanism can strongly implement the first best if and only if $c_2^*/R > c_1^*/R^a$. The proof of Theorem 1 demonstrates that this is the case because the asset transfer mechanism can strongly implement the first best if and only if there is an asset transfer $\kappa$ that 1) requires less asset than must be used to supply a depositor with $c_2^*$ goods in period 2 ($\kappa \leq c_2^*/R$); and 2) is strictly preferred by patient depositors over withdrawing $c_1^*$ goods in period 1 ($R^a\kappa > c_1^*$). Translating this logic into our setting with idiosyncratic return risk, 1) becomes $\kappa \leq c_2^*/\bar{R}$, and 2) becomes

$$qu(R_H\kappa) + (1-q)u(R_L\kappa) > u(c_1^*).$$

Since the left-hand side is strictly increasing in $\kappa$, it follows that the asset transfer mechanism can strongly implement the first best if and only if the inequality holds at $\kappa = c_2^*/\bar{R}$:

$$qu(R_H c_2^*/\bar{R}) + (1-q)u(R_L c_2^*/\bar{R}) > u(c_1^*).$$

We now show that there exists a $\bar{\eta} \in (1,\infty)$ such that this inequality holds if and only if $\eta < \bar{\eta}$, where $\eta \equiv R_H/R_L$. Note that if we vary $\eta$ while keeping $\bar{R}$ and $q$ fixed, then $c_1^*$ is constant.

Using $R_H = \eta R_L$ and $\bar{R} = (q\eta + 1 - q)R_L$, we can write

$$qu(R_H c_2^*/\bar{R}) + (1-q)u(R_L c_2^*/\bar{R}) = qu\left(\frac{\eta c_2^*}{q\eta+1-q}\right) + (1-q)u\left(\frac{c_2^*}{q\eta+1-q}\right),$$

which we call $U(\eta)$ because we are varying $\eta$ while keeping $\bar{R}$ (and so $c_2^*$) and $q$ fixed. Since $U(\cdot)$ is differentiable, it is sufficient to show that $U(\eta)$ is strictly decreasing in $\eta$, that $U(1) > u(c_1^*)$, and that $\lim_{\eta\to\infty} U(\eta) < u(c_1^*)$. First,

$$U'(\eta) = \frac{q(1-q)c_2^*}{(q\eta+1-q)^2}\left(u'\left(\frac{\eta c_2^*}{q\eta+1-q}\right) - u'\left(\frac{c_2^*}{q\eta+1-q}\right)\right),$$

which is equal to 0 at $\eta = 1$, and strictly negative for all $\eta > 1$ because $u$ is strictly concave. Hence, $U(\eta)$ is strictly decreasing in $\eta$. Second, $U(1) = u(c_2^*)$, which is strictly greater than $u(c_1^*)$. Finally, as $\eta$ goes to infinity, $U(\eta)$ goes to negative infinity (strictly less than the $u(c_1^*)$) because the argument of $u$ in the second term goes to 0, and we made the additional technical assumption on preferences in Section 3.3.1 that $\lim_{c\to 0} u(c) = -\infty$. □

# E Sophisticated Depositors (For Online Publication)

In this appendix, we consider an alternative version of our idiosyncratic risk environment with "sophisticated depositors" that can observe the return on assets they are offered for transfer. As in Section 3.3, investment occurs through risky projects: each investment must be at least $\bar{x}$ units of the asset, and the return on an investment is $R_H$ (high quality) with probability $q$ and $R_L \leq R_H$

(low quality) with probability $1-q$. The average return is $\bar{R} = (1-q)R_L + qR_H$. Investments are indivisible, so they must be transferred in their entirety and liquidated or held to maturity in their entirety.

The difference from Section 3.3 is that if the intermediary offers a depositor an asset transfer, then the depositor can observe whether the assets offered are high or low quality before they choose their action (withdraw goods, wait, or take the asset transfer). We call a depositor offered high quality assets an $H$-type, and a depositor offered low quality assets an $L$-type. The intermediary cannot observe whether individual investments are high or low quality, but in equilibrium, uses Bayes' rule to learn about the returns on particular investments by observing the actions of depositors. As such, the intermediary now must choose a "protocol" for which assets it will liquidate to provide goods and which assets it will offer as transfers to depositors. The intermediary can use assets that have not previously been offered, which are high quality with probability $q$, or the intermediary can use assets that have been rejected by potentially many depositors at various places in line, which can be high quality with probability not equal to $q$. Depositors then understand that the share of depositors offered high quality asset transfers may differ from the ex-ante probability $q$, and the average return on the intermediary's assets at the end of period 1 may differ from the ex-ante average $\bar{R}$.

Before we turn to the effectiveness of the asset transfer mechanism, we note that the constrained planner may now be able to use sophisticated depositors' information to offer a better allocation than it could without idiosyncratic return risk. However, this requires using asset transfers in equilibrium, and so is not possible with the demand deposit mechanism. Thus for consistency with the rest of our results, we restrict attention to allocations of the form $(\bar{c}_1, \bar{c}_2)$ that are feasible without depositor information, i.e., $(1-\lambda)\bar{c}_2 = \bar{R}(1-\lambda\bar{c}_1)$, and that provide insurance, i.e., $1 < \bar{c}_1 < \bar{c}_2 < \bar{R}$. We then ask whether the asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$, i.e., an allocation in which each impatient depositor consumes at least $\bar{c}_1$ and each patient depositor consumes at least $\bar{c}_2$.

For simplicity, as in our original Section 3.3 on idiosyncratic return risk, we suppose parameters are such that it is sufficient to consider the case in which the intermediary offers as its asset transfer one investment of size $\kappa$ to each depositor and, when choosing $\kappa$, the constraint $\kappa \geq \bar{x}$ does not bind. Thus, the intermediary's asset holdings at the beginning of period 1 are a measure $1/\kappa$ of indivisible investments, each of size $\kappa$. Each investment has return $R_H$ with probability $q$ and return $R_L$ with probability $1-q$. Moreover, we suppose asset transfers are frictionless, i.e., $R^a = R$ or equivalently, $\varepsilon = 0$. Finally, we impose on preferences that $\lim_{c\to 0} u(c) = -\infty$.

Our main result is the following proposition, which characterizes when the asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$.

**Proposition 7.** *Suppose all depositors are "sophisticated" and the allocation $(\bar{c}_1, \bar{c}_2)$ is feasible without depositor information and provides insurance. Vary return dispersion, $\eta \equiv R_H/R_L$, and the probability of a high return, $q$, while adjusting the return realizations $R_H$ and $R_L$ so that the average return, $\bar{R}$, is constant.*

*The asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ if and only if $q < q_1(\eta)$ or $q > q_2(\eta)$, where*

1. *$q_1(\cdot)$ and $q_2(\cdot)$ are continuous and strictly decreasing;*
2. *if $\eta \leq \bar{c}_2/\bar{c}_1$, then $q_1(\eta) \geq 1$ so that strong implementation is possible for any $q$;*

3. *for sufficiently large $\eta$, $0 < q_1(\eta) < q_2(\eta) < 1$, so that strong implementation is not possible for $q$ in the non-empty interval $[q_1(\eta), q_2(\eta)]$.*

Before proving the proposition, we briefly discuss the result and provide intuition. The proposition states that with sophisticated depositors, there are two ways for the asset transfer mechanism to strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$. The first way works if return dispersion, $\eta$, and the probability of a high quality return, $q$, are sufficiently low, i.e., if $q < q_1(\eta)$ where $q_1(\eta)$ is strictly decreasing. The second way works if $\eta$ and $q$ are sufficiently high, i.e., if $q > q_2(\eta)$ where $q_2(\eta)$ is strictly decreasing. Moreover if $\eta \leq \bar{c}_2/\bar{c}_1$, then the first way works for any value of $q$. By contrast, if $\eta$ is sufficiently large, then there are values of $q$ at which neither way works.

More specifically, the first way to strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ is to offer an asset transfer that all patient depositors prefer over withdrawing goods early. Such an asset transfer is effective if it costs less than $\bar{c}_2/\bar{R}$, which is the assets a patient depositor uses if they wait and get $\bar{c}_2$. This is possible if the low quality return, $R_L$, is sufficiently close to the average return, $\bar{R}$. In turn, this holds if return dispersion, $\eta$, and the probability of a high return, $q$, are sufficiently low because lowering $\eta$ compresses $R_L$ and $R_H$ toward $\bar{R}$, and lowering $q$ increases the importance of the low return to the average. Thus, the asset transfer mechanism can eliminate runs if $q < q_1(\eta)$, where $q_1(\eta)$ is strictly decreasing. Note that this is exactly as in Theorem 1 and Corollary 1 in our baseline environment, but with the deterministic return $R$ replaced by the average return $\bar{R}$ and with the depositor return $R^a$ replaced by the low quality return $R_L$.

Second, the intermediary can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ by offering an asset transfer that patient depositors prefer to withdrawing goods early only if they are $H$-type, i.e., only if they are offered high quality assets with return $R_H$. Such an asset transfer is effective if return dispersion, $\eta$, and the probability of a high return, $q$, are sufficiently big. This is because an increase in $\eta$ pushes $R_L$ and $R_H$ away from $\bar{R}$, so it is cheaper to provide an asset transfer that patient $H$-type depositors prefer over withdrawing goods early. Moreover, an increase in $q$ means a larger share of patient depositors are willing to take the asset transfer. Thus, the asset transfer mechanism can strongly implement if $q > q_2(\eta)$, where $q_2(\eta)$ is strictly decreasing.

In the second case, there is the concern that if patient $H$-type depositors take their high quality asset transfers, then the pool of remaining assets is disproportionately low quality, which encourages other patient depositors to withdraw goods early rather than wait. We eliminate this problem by using what we call the "standard protocol" to select assets to offer as transfers and to liquidate for providing goods. Specifically, if a depositor withdraws goods, then their offered transfer is liquidated, and if a depositor waits, then their offered transfer is offered to the next depositor. This eliminates the problem because if patient depositors withdraw goods early conditional on being offered low quality assets, then their low quality asset transfers are liquidated, which removes them from the intermediary's balance sheet.

*Proof of Proposition 7.* The proof proceeds in several steps. First, we describe the "standard protocol" and compute the implied share of patient depositors that are $H$-type (offered a high quality asset transfer), as well as the implied average return on the intermediary's assets at the end of period 1. Second, we show that the asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ if $\bar{c}_1/R_L < \bar{c}_2/\bar{R}$. Third, we show that there is a continuous, strictly decreasing function $q_1(\eta) \in (0,1)$ such that $\bar{c}_1/R_L < \bar{c}_2/\bar{R}$ if and only if $q < q_1(\eta)$. Fourth, we state an inequality such that the asset transfer mechanism can strongly implement an allocation that

weakly dominates $(\bar{c}_1, \bar{c}_2)$ if the inequality holds. Fifth, we show that there is a continuous, strictly decreasing function $q_2(\eta) \in (0,1)$ such that the inequality holds if and only if $q > q_2(\eta)$. Sixth, we show that if $q_1(\eta) \leq q \leq q_2(\eta)$, then the asset transfer mechanism cannot strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$.

Step 1: The intermediary's protocol outlines which assets it will liquidate to provide goods and which assets it will offer as transfers to depositors. Define the "standard protocol" as follows, taking as given that the asset transfer is smaller than the goods withdrawal. If a depositor waits, then offer their asset transfer to the next depositor. If a depositor withdraws goods, then to provide them goods, liquidate the offered asset transfer in its entirety as well as other assets that have not been previously offered (recall, the asset transfer is smaller than the goods withdrawal). If a depositor withdraws goods or takes the asset transfer, then for the next depositor's asset transfer, offer assets that have not been previously offered.

We now suppose the intermediary uses the standard protocol and compute the implied share of patient depositors that are $H$-type. Suppose impatient depositors never wait, patient $L$-type depositors wait with probability $x_L$, and patient $H$-type depositors wait with probability $x_H$. The share of patient depositors that are $H$-type depends on how many times an asset transfer is reused if it is low quality vs. if it is high quality. Suppose the intermediary uses a measure $z$ of unique investments, each of size $\kappa$, as asset transfers throughout period 1. Since an asset transfer must initially be drawn from the pool of assets that have not previously been offered, it follows that a fraction $q$ of these investments are high quality. A high quality investment is offered to $n$ depositors as a transfer with probability $(1-\lambda)x_H^{n-1}(1-(1-\lambda)x_H)$ because each time it is offered, it is offered again if the depositor is patient and waits, which occurs with probability $(1-\lambda)x_H$. As such, the expected number of depositors to which a given high quality investment is offered is

$$\sum_{n=1}^{\infty} n((1-\lambda)x_H)^{n-1}(1-(1-\lambda)x_H) = \left(\sum_{n=1}^{\infty}((1-\lambda)x_H)^{n-1}\right)^2 (1-(1-\lambda)x_H) = \frac{1}{1-(1-\lambda)x_H}.$$

Therefore, the measure $qz$ of high quality investments are offered to a measure $qz/(1-(1-\lambda)x_H)$ of depositors and analogously, the measure $(1-q)z$ of low quality investments are offered to a measure $(1-q)z/(1-(1-\lambda)x_L)$ of depositors. Thus, the share of patient depositors that are $H$-type (offered a high quality asset transfer) is

$$\frac{qz/(1-(1-\lambda)x_H)}{qz/(1-(1-\lambda)x_H)+(1-q)z/(1-(1-\lambda)x_L)} = \frac{q}{q+(1-q)\frac{1-(1-\lambda)x_H}{1-(1-\lambda)x_L}}. \tag{E.1}$$

Next, we show that if the intermediary uses the standard protocol, then the average return on the intermediary's assets at the end of period 1 is the ex-ante average return, $\bar{R}$. To see this, note that if a depositor rejects their asset transfer, then either the transfer is liquidated or offered to the next depositor. Hence, all but a measure 0 of the intermediary's assets at the end of period 1 have not been previously offered. It follows that the intermediary's assets at the end of period 1 are high quality with the ex-ante probability $q$, so they have average return $\bar{R}$.

Step 2: We show that the asset transfer mechanism can strongly implement an allocation that

weakly dominates $(\bar{c}_1, \bar{c}_2)$ if $\bar{c}_1/R_L < \bar{c}_2/\bar{R}$. Specifically, set the goods withdrawal to $\bar{c}_1$, set the asset transfer to $\kappa \in (\bar{c}_1/R_L, \bar{c}_2/\bar{R})$, and use the standard protocol described in Step 1 to select assets to offer as transfers and to liquidate for providing goods.

We show that in any equilibrium, the allocation weakly dominates $(\bar{c}_1, \bar{c}_2)$. The proof follows the same argument as the proof of Theorem 1. First, since the intermediary uses the standard protocol, Step 1 tells us that the average return on its assets at the end of period 1 is $\bar{R}$. Second, the asset transfer is high enough so that all patient depositors strictly prefer taking the transfer over withdrawing goods because holding the transfer to maturity yields $R_L\kappa > \bar{c}_1$ goods for an $L$-type depositor and $R_H\kappa > R_L\kappa > \bar{c}_1$ goods for an $H$-type depositor. Third, since $\kappa < \bar{c}_2/\bar{R}$, the asset transfer uses fewer assets than patient depositors use if everyone's strategy is "truth-telling" (impatient depositors withdraw goods, patient depositors wait, and the allocation is $(\bar{c}_1, \bar{c}_2)$). Therefore, as in the proof of Theorem 1, it follows that in any equilibrium, impatient depositors withdraw goods, patient depositors mix between taking the asset transfer and waiting, and a depositor who waits gets at least $\bar{c}_2$ goods in period 2. So in any equilibrium, impatient depositors consume $\bar{c}_1$ and patient depositors consume at least $\bar{c}_2$ because they can get at least $\bar{c}_2$ by waiting.

Step 3: We show that there exists a continuous, strictly decreasing function $q_1(\eta) > 0$ such that $\bar{c}_1/R_L < \bar{c}_2/\bar{R}$ if and only if $q < q_1(\eta)$. Moreover, we show that $q_1(\eta) \geq 1$ if $\eta \leq \bar{c}_2/\bar{c}_1 - 1$. Use $\eta \equiv R_H/R_L$ to write the ex-ante average asset return as

$$\bar{R} = (1-q)R_L + qR_H = ((1-q) + q\eta)R_L.$$

Plugging in, we get that $\bar{c}_1/R_L < \bar{c}_2/\bar{R}$ if and only if $(1-q) + q\eta < \bar{c}_2/\bar{c}_1$, which holds if and only if $q < q_1(\eta)$, where

$$q_1(\eta) = \frac{\bar{c}_2/\bar{c}_1 - 1}{\eta - 1}.$$

$q_1(\eta)$ is continuous and strictly decreasing in $\eta$. Moreover, $q_1(\eta) > 0$ because $\bar{c}_2 > \bar{c}_1$ and $\eta > 1$. Finally, $q_1(\bar{c}_2/\bar{c}_1) = 1$ and $q_1(\eta)$ is decreasing in $\eta$, so it follows that if $\eta \leq \bar{c}_2/\bar{c}_1$, then $q_1(\eta) \geq 1$.

Step 4: We show that the asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ if

$$1 - \lambda\bar{c}_1 - (1-\lambda)(1-q)\bar{c}_1 - (1-\lambda)q\bar{c}_1/R_H > 0. \tag{E.2}$$

Specifically, use the standard protocol described in Step 1 to select assets to liquidate for providing goods and to offer as transfers. Set the goods withdrawal to $\bar{c}_1$. Set the asset transfer to $\kappa > \bar{c}_1/R_H$ such that the following hold. First, $\kappa < \min\{\bar{c}_2/R_H, \bar{c}_1/R_L, \bar{c}_1\}$, which is possible given $\kappa > \bar{c}_1/R_H$ because $\bar{c}_2 < \bar{c}_1$, $R_H > R_L$, and $R_H > 1$. Second, $\kappa$ is low enough so that

$$1 - \lambda\bar{c}_1 - (1-\lambda)(1-q)\bar{c}_1 - (1-\lambda)q\kappa > 0, \tag{E.3}$$

which is possible given $\kappa > \bar{c}_1/R_H$ because inequality (E.2) holds.

First, we eliminate strictly dominated strategies. Impatient depositors strictly prefer withdrawing goods over taking the asset transfer because the former yields $\bar{c}_1$ goods, whereas the latter yields $\kappa < \bar{c}_1$ by construction. Moreover, impatient depositors strictly prefer withdrawing goods

over waiting because they do not value goods in period 2. Patient $L$-type depositors also strictly prefer withdrawing goods over taking the asset transfer because for them, holding the asset transfer to maturity yields $R_L\kappa$ goods, which is strictly less than $\bar{c}_1$ by construction. Finally, patient $H$-type depositors strictly prefer taking the asset transfer over withdrawing goods because for them, holding the asset transfer to maturity yields $R_H\kappa$ goods, which is strictly greater than $\bar{c}_1$ by construction.

Now in any equilibrium, depositors do not play strictly dominated strategies, so impatient depositors withdraw goods, patient $L$-type depositors withdraw goods with probability $\alpha$ and otherwise wait, and patient $H$-type depositors take the asset transfer with probability $\beta$ and otherwise wait. We split potential equilibria into three cases. In each case, we show that a depositor who waits get at least $\bar{c}_2$ goods in period 2. It follows that in any equilibrium, all patient depositors wait because withdrawing goods yields $\bar{c}_1 < \bar{c}_2$ goods and holding a high quality asset transfer to maturity yields $R_H\kappa < \bar{c}_2$ goods by construction. This then implies that there is a unique equilibrium: impatient depositors withdraw goods, patient depositors wait, and the allocation is $(\bar{c}_1, \bar{c}_2)$.

For the first case of a potential equilibrium, suppose patient $H$-type depositors wait with strictly positive probability ($\beta < 1$). Since patient $H$-type depositors sometimes wait, it follows that waiting yields weakly more period 2 goods than a patient $H$-type depositor's best alternative, which is $R_H\kappa$ from taking the asset transfer. Then patient $L$-type depositors must wait with probability one ($\alpha = 0$) because their best alternative to waiting is withdrawing goods, which is strictly worse than taking a high quality asset transfer, i.e., it yields only $\bar{c}_1 < R_H\kappa$. As such, a depositor who waits gets period 2 consumption

$$\bar{R}\frac{\max\{1-\lambda\bar{c}_1-(1-\lambda)\tilde{q}\beta\kappa,0\}}{(1-\lambda)(1-\tilde{q}+\tilde{q}(1-\beta))},$$

where the numerator is the measure of assets left on the intermediary's balance sheet at the end of period 1, the denominator is the measure of depositors who wait, $\tilde{q}$ is the share of patient depositors that are $H$-type given by (E.1) in Step 1:

$$\tilde{q} = \frac{q}{q+(1-q)\frac{\lambda+(1-\lambda)\beta}{\lambda}},$$

and $\bar{R}$ is the average return on the intermediary's assets remaining at the end of period 1 (by Step 1 since we use the standard protocol). Algebra then yields that the consumption from waiting is

$$\bar{R}\frac{\max\{1-\lambda\bar{c}_1-(1-\lambda)\tilde{q}\beta\kappa,0\}}{(1-\lambda)(1-\tilde{q}+\tilde{q}(1-\beta))} = \frac{\bar{c}_2-\tilde{q}\beta\bar{R}\kappa}{1-\tilde{q}\beta} \geq \bar{c}_2,$$

where the equality uses that feasibility of $(\bar{c}_1, \bar{c}_2)$ means $(1-\lambda)\bar{c}_2 = \bar{R}(1-\lambda\bar{c}_1)$, and the inequality uses $\kappa < \bar{c}_2/\bar{R}$. Therefore, if there is an equilibrium in which patient $H$-type depositors wait with strictly positive probability ($\beta < 1$), then in this equilibrium, a depositor who waits consumes at least $\bar{c}_2$.

For the second case of a potential equilibrium, suppose patient $L$-type depositors always withdraw goods and patient $H$-type depositors always take the asset transfer ($\alpha = \beta = 1$). In this case, the measure of assets remaining on the intermediary's balance sheet at the end of period 1 is

$$\max\{1-\lambda\bar{c}_1-(1-\lambda)(1-q)\bar{c}_1-(1-\lambda)q\kappa,0\},$$

where we use (E.1) in Step 1 to get that with $\alpha = \beta = 1$, the share of patient depositors that are $H$-type is $q$. The first argument in the maximum is the left-hand side of (E.3), so it is strictly positive by construction. Hence, the intermediary has goods to give out in period 2, but a measure 0 of depositors wait. Thus, if there is an equilibrium in which patient $L$-type depositors always withdraw goods and patient $H$-type depositors always wait, then in this equilibrium, a depositor who waits gets infinite goods.

For the last case of a potential equilibrium, suppose patient $L$-type depositors wait with strictly positive probability and patient $H$-type depositors always take the asset transfer ($\alpha < 1$ and $\beta = 1$). In this case, a depositor who waits gets period 2 consumption

$$c_2(\alpha) = \bar{R}\frac{\max\{1-(\lambda+(1-\lambda)(1-\tilde{q})\alpha)\bar{c}_1-(1-\lambda)\tilde{q}\kappa,0\}}{(1-\lambda)(1-\tilde{q})(1-\alpha)},$$

where the numerator is the measure of assets left on the intermediary's balance sheet at the end of period 1, the denominator is the measure of depositors who wait, $\tilde{q}$ is the share of patient depositors that are $H$-type given by (E.1) in Step 1:

$$\tilde{q} = \frac{q}{q+(1-q)\frac{1}{\lambda+(1-\lambda)\alpha}},$$

and $\bar{R}$ is the average return on the intermediary's assets remaining at the end of period 1 (by Step 1 since we use the standard protocol). The first argument in the maximum in the numerator is decreasing in $\alpha$ (including the effect on $\tilde{q}$) because

$$\frac{\partial\tilde{q}}{\partial\alpha} = \tilde{q}(1-\tilde{q})\frac{1-\lambda}{\lambda+(1-\lambda)\alpha} > 0 \qquad \text{and} \qquad \frac{\partial(1-\tilde{q})\alpha}{\partial\alpha} = (1-\tilde{q})\frac{\lambda+(1-\lambda)\alpha(1-\tilde{q})}{\lambda+(1-\lambda)\alpha} > 0.$$

Hence, the first argument in the maximum is strictly positive because we saw earlier that it is strictly positive at $\alpha = \beta = 1$. Then, the derivative of a waiting depositor's period 2 consumption with respect to $\alpha$ is

$$\begin{aligned}\frac{\partial c_2(\alpha)}{\partial\alpha} =& -\bar{R}\frac{(1-\lambda)(1-\tilde{q})\frac{\lambda+(1-\lambda)\alpha(1-\tilde{q})}{\lambda+(1-\lambda)\alpha}\bar{c}_1+(1-\lambda)\tilde{q}(1-\tilde{q})\frac{1-\lambda}{\lambda+(1-\lambda)\alpha}\kappa}{(1-\lambda)(1-\tilde{q})(1-\alpha)}\\ &+\frac{c_2(\alpha)}{(1-\tilde{q})(1-\alpha)}\left((1-\tilde{q})\frac{\lambda+(1-\lambda)\alpha(1-\tilde{q})}{\lambda+(1-\lambda)\alpha}+\tilde{q}(1-\tilde{q})\frac{1-\lambda}{\lambda+(1-\lambda)\alpha}\right)\\ =&\frac{1}{(\lambda+(1-\lambda)\alpha)(1-\alpha)}F(\alpha),\end{aligned}$$

where

$$F(\alpha) = -(\lambda+(1-\lambda)\alpha(1-\tilde{q}))\bar{R}\bar{c}_1-(1-\lambda)\tilde{q}\bar{R}\kappa+(\lambda+(1-\lambda)\alpha(1-\tilde{q})+(1-\lambda)\tilde{q})c_2(\alpha).$$

The derivative of $F(\alpha)$ is

$$\begin{aligned}\frac{\partial F(\alpha)}{\partial \alpha} =& -(1-\lambda)(1-\tilde{q})\frac{\lambda+(1-\lambda)\alpha(1-\tilde{q})}{\lambda+(1-\lambda)\alpha}\bar{R}\bar{c}_1-(1-\lambda)\tilde{q}(1-\tilde{q})\frac{1-\lambda}{\lambda+(1-\lambda)\alpha}\bar{R}\kappa\\ &+(1-\lambda)(1-\tilde{q})\frac{\lambda+(1-\lambda)\alpha(1-\tilde{q})}{\lambda+(1-\lambda)\alpha}c_2(\alpha)+(1-\lambda)\tilde{q}(1-\tilde{q})\frac{1-\lambda}{\lambda+(1-\lambda)\alpha}c_2(\alpha)\\ &+(\lambda+(1-\lambda)\alpha(1-\tilde{q})+(1-\lambda)\tilde{q})\frac{\partial c_2(\alpha)}{\partial \alpha}\\ =&\frac{(1-\lambda)(1-\tilde{q})}{\lambda+(1-\lambda)\alpha}F(\alpha)+(\lambda+(1-\lambda)\alpha(1-\tilde{q})+(1-\lambda)\tilde{q})\frac{\partial c_2(\alpha)}{\partial \alpha}\\ =&\frac{1}{(\lambda+(1-\lambda)\alpha)(1-\alpha)}F(\alpha).\end{aligned}$$

Therefore, $\partial F(\alpha)/\partial\alpha$ is positive if and only if $F(\alpha)$ is positive. It follows that if $F(0)\geq 0$, then for all $\alpha\in[0,1)$, $F(\alpha)\geq 0$, and if $F(0)\leq 0$, then for all $\alpha\in[0,1)$, $F(\alpha)\leq 0$. Thus, $c_2(\alpha)$ is monotone in $\alpha$. Furthermore, $c_2(\alpha)$ is increasing in $\alpha$ because we saw earlier that as $\alpha$ goes to 1, $c_2(\alpha)$ diverges to positive infinity. So for all $\alpha\in[0,1)$, $c_2(\alpha)\geq c_2(0)$, which is given by

$$c_2(0)=\bar{R}\frac{1-\lambda\bar{c}_1-(1-\lambda)\tilde{q}\kappa}{(1-\lambda)(1-\tilde{q})}=\frac{\bar{c}_2-\tilde{q}\bar{R}\kappa}{1-\tilde{q}}>\bar{c}_2,$$

where the second equality uses that feasibility of $(\bar{c}_1,\bar{c}_2)$ means $(1-\lambda)\bar{c}_2=\bar{R}(1-\lambda\bar{c}_1)$, and the inequality uses $\kappa<\bar{c}_2/\bar{R}$. Therefore, if there is an equilibrium in which patient $L$-type depositors wait with strictly positive probability and patient $H$-type depositors always take the asset transfer ($\alpha<1$ and $\beta=1$), then in this equilibrium, a depositor who waits gets at least $\bar{c}_2$ goods. This completes the proof of Step 4.

Step 5: We show that there exists a continuous, strictly decreasing function $q_2(\eta)\in(0,1)$ such that inequality (E.2) holds if and only if $q>q_2(\eta)$. Moreover, we show that for sufficiently large $\eta$, $q_2(\eta)>q_1(\eta)$. Use the definitions of the average return, $\bar{R}=(1-q)R_L+qR_H$, and of return dispersion, $\eta=R_H/R_L$, to get that $\bar{R}/R_H=(1-q)/\eta+q$. Use this to plug in for $R_H$ in the left-hand side of inequality (E.2):

$$\begin{aligned}&1-\lambda\bar{c}_1-(1-\lambda)(1-q)\bar{c}_1-(1-\lambda)q\bar{c}_1/R_H\\ &=1-\lambda\bar{c}_1-(1-\lambda)(1-q)\bar{c}_1-(1-\lambda)q(1-q)\frac{\bar{c}_1}{\eta\bar{R}}-(1-\lambda)q^2\frac{\bar{c}_1}{\bar{R}}\equiv G(q).\end{aligned}$$

So inequality (E.2) holds if and only if $G(q)>0$, where $G(q)$ is a second-order polynomial in $q$. At $q=0$, the polynomial is $G(0)=1-\bar{c}_1$, which is strictly negative because $(\bar{c}_1,\bar{c}_2)$ provides insurance. As $q$ goes to positive infinity, $G(q)$ goes to negative infinity. Finally, at $q=1$,

$$G(1)=1-\lambda\bar{c}_1-(1-\lambda)\frac{\bar{c}_1}{\bar{R}},$$

which is strictly positive because feasibility of $(\bar{c}_1, \bar{c}_2)$ means that $1 - \lambda\bar{c}_1 = (1-\lambda)\bar{c}_2/\bar{R}$, and because $\bar{c}_2 > \bar{c}_1$. Hence, the polynomial $G(q)$ has two zeros, $z_1 < z_2$, where $z_1 \in (0,1)$ and $z_2 > 1$: the polynomial is negative for $q < z_1$, positive for $q \in (z_1, z_2)$, and negative for $q > z_2$.

Now, define $q_2(\eta)$ to be the lower root, $z_1$, which is in $(0,1)$. Then since $q$ must be in $(0,1)$ and since inequality (E.2) holds if and only if $G(q) > 0$, it follows that inequality (E.2) holds if and only if $q > q_2(\eta)$. Moreover, $G(q)$ is strictly increasing in $\eta$ for $q \in (0,1)$, so a higher $\eta$ means that if the polynomial used to be 0, then it is now strictly positive. Therefore the lower root, $q_2(\eta)$, must be strictly decreasing in $\eta$. We can also see that $q_2(\eta)$ is continuous because the polynomial $G(q)$ is continuous in $\eta$. Finally, as $\eta$ goes to infinity, $q_2(\eta)$ is bounded away from 0 because $G(q)$ converges to the polynomial

$$1 - \lambda\bar{c}_1 - (1-\lambda)(1-q)\bar{c}_1 - (1-\lambda)q^2\frac{\bar{c}_1}{\bar{R}},$$

which has a lower root strictly between 0 and 1, just like with finite $\eta$. Then since $q_1(\eta)$ goes to 0 as $\eta$ goes to infinity, it follows that for sufficiently large $\eta$, $q_2(\eta) > q_1(\eta)$. This completes the proof of Step 5.

Step 6: In this step, we complete the proof of the proposition. We constructed functions $q_1(\eta)$ (Step 3) and $q_2(\eta)$ (Step 5) such that if $q < q_1(\eta)$ or if $q > q_2(\eta)$, then the asset transfer mechanism can strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$. We now show the opposite direction: if $q_1(\eta) \leq q \leq q_2(\eta)$, then the asset transfer mechanism cannot strongly implement an allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$. Suppose $q_1(\eta) \leq q \leq q_2(\eta)$. Since $q \geq q_1(\eta)$, then by construction, $\bar{c}_1/R_L \geq \bar{c}_2/\bar{R}$. Since $q \leq q_2(\eta)$, then by construction, inequality (E.2) does not hold.

We show that for any value of the asset transfer $\kappa$, there is an equilibrium allocation that does not weakly dominate $(\bar{c}_1, \bar{c}_2)$. In particular, we show this is the case under any protocol for selecting assets to offer as transfers and to liquidate for providing goods. First, suppose the asset transfer is $\kappa \geq \bar{c}_1$. Then there is an equilibrium in which depositors choose whichever they prefer between withdrawing goods and taking the asset transfer. To see this, note that depositors only choose to withdraw goods if the goods withdrawal is at least $\kappa$, which is weakly greater than $\bar{c}_1$. So each depositor uses at least $\bar{c}_1$ assets in period 1. We know that $\bar{c}_1 > 1$ because $(\bar{c}_1, \bar{c}_2)$ provides insurance, so it follows that the intermediary runs out of assets before the end of period 1. Hence, a depositor who waits gets nothing. Therefore, the proposed strategies form an equilibrium. The equilibrium allocation does not weakly dominate $(\bar{c}_1, \bar{c}_2)$ because depositors who arrive after the intermediary runs out of assets consume nothing.

For the rest of the proof, suppose $\kappa < \bar{c}_1$. Then the goods withdrawal must be at least $\bar{c}_1$ to get an equilibrium allocation that weakly dominates $(\bar{c}_1, \bar{c}_2)$ because otherwise impatient depositors consume strictly less than $\bar{c}_1$. It follows that impatient depositors strictly prefer to withdraw goods over taking the asset transfer. Now if $\bar{c}_2/\bar{R} < \kappa < \bar{c}_1$, then there is an equilibrium in which impatient depositors withdraw goods and patient depositors choose whichever they prefer between withdrawing goods and taking the asset transfer. To see this, note that whether patient depositors withdraw goods or take the asset transfer, they use at least $\kappa > \bar{c}_2/\bar{R}$ assets in period 1. Hence, the intermediary runs out of assets before the end of period 1 because feasibility of $(\bar{c}_1, \bar{c}_2)$ means the intermediary has exactly enough assets for each impatient depositor to use $\bar{c}_1$ and for each patient depositor to use $\bar{c}_2/\bar{R}$. Therefore, the proposed strategies form an equilibrium because a depositor

who waits gets nothing. The equilibrium allocation does not weakly dominate $(\bar{c}_1, \bar{c}_2)$ because depositors who arrive after the intermediary runs out of assets consume nothing.

Next, if $\bar{c}_1/R_H < \kappa \leq \bar{c}_2/\bar{R}$ and the goods withdrawal is weakly less than $R_H\kappa$ (but still weakly greater than $\bar{c}_1$), then there is an equilibrium in which impatient and patient $L$-type depositors withdraw goods and patient $H$-type depositors take the asset transfer. To see this, note that patient $L$-type depositors prefer to withdraw goods over taking the asset transfer because for them, the asset transfer yields $R_L\kappa$ goods, which is weakly less than $\bar{c}_2/\bar{R}$ by assumption, which is strictly less than $\bar{c}_1$ because $(\bar{c}_1, \bar{c}_2)$ provides insurance. On the other hand, patient $H$-type depositors prefer to take the asset transfer over withdrawing goods because for them, the asset transfer yields $R_H\kappa$ goods, which is greater than $\bar{c}_1$ by construction. Hence, the measure of assets left on the intermediary's balance sheet at the end of period 1 is

$$\max\{1 - \lambda\bar{c}_1 - (1-\lambda)(1-\tilde{q})\bar{c}_1 - (1-\lambda)\tilde{q}\kappa, 0\}, \tag{E.4}$$

where $\tilde{q}$ is the share of patient depositors that are $H$-type. We want to show that this maximum is 0 for any asset transfer $\kappa$ such that $\bar{c}_1/R_H < \kappa \leq \bar{c}_2/\bar{R}$, and for any protocol. The first argument in the maximum is decreasing in $\kappa$, which is strictly greater than $\bar{c}_1/R_H$. Moreover, the first argument in the maximum is increasing in $\tilde{q}$ because $\kappa < \bar{c}_1$. To find the highest value of $\tilde{q}$, note that under any protocol, offered units of the asset must be units that were not previously offered or units that were previously rejected. The former are high quality with the ex-ante probability $q$, whereas the latter are high quality with probability less than $q$ because patient depositors only reject low quality assets and impatient depositors reject all assets. As such, $\tilde{q} \leq q$. It follows that the first argument in the maximum in (E.4) is strictly less than its value with $\kappa = \bar{c}_1/R_H$ and with $\tilde{q} = q$, which is the left-hand side of (E.2), which is weakly negative by assumption. Thus, the intermediary runs out of assets before the end of period 1. Therefore, the proposed strategies form an equilibrium because a depositor who waits gets nothing. The equilibrium allocation does not weakly dominate $(\bar{c}_1, \bar{c}_2)$ because depositors who arrive after the intermediary runs out of assets consume nothing.

Finally, if $\kappa \leq \bar{c}_1/R_H$, or if $\bar{c}_1/R_H < \kappa \leq \bar{c}_2/\bar{R}$ and the goods withdrawal is weakly greater than $R_H\kappa$, then there is a run equilibrium in which all depositors withdraw goods. To see this, note that patient $H$-type depositors prefer withdrawing goods over taking the asset transfer because for them, the asset transfer yields $R_H\kappa$ goods, which is less than the goods withdrawal by construction. It follows that all depositors prefer withdrawing goods over taking the asset transfer because patient $H$-type depositors get the most value from taking the asset transfer. Now if all depositors withdraw goods, then the intermediary runs out of assets before the end of period 1 because the goods withdrawal is at least $\bar{c}_1$, which is strictly greater than 1 since $(\bar{c}_1, \bar{c}_2)$ provides insurance. Hence, the proposed strategies form an equilibrium because a depositor who waits gets nothing. The equilibrium allocation does not weakly dominate $(\bar{c}_1, \bar{c}_2)$ because depositors who arrive after the intermediary runs out of assets consume nothing. This completes the proof of Step 6.

□